SUPPLEMENTARY MATERIAL

References and citation numbers in the supplementary material follow the bibliography of the main paper.

S.1. DATASETS

Table II in the main text summarises dataset structure and the complete downstream target matrix. Table S1 complements it with class balances, anomaly rates, and the horizon used for every target. Tables S2–S4 report statistics of the released representation, exact model inputs, and the effective number of downstream examples.

We distinguish targets by their scope independently of the downstream task family. A global target assigns one value to an entire entity sequence and remains unchanged across its sampled slices. A local target is defined separately at each sampled cutoff using the subsequent dataset-specific horizon; different slices of the same sequence may therefore have different target values. All forecasting targets and all but one regression target are local, whereas classification and anomaly detection include both global and local targets. Table S1 reports the scope and horizon of every target.

S.1.1 Dataset Sources

The transaction datasets are AlphaBattle[2], AGE[3], and Retail[4]. The recommender-system datasets are Zvuk[5], Yambda[6], 30Music[7], and Taobao[8]. The time-series datasets are Electric Devices[9], ETT[10], Favorita[11], and Rossmann[12]. The text dataset is Twitter Sentiment[13].

S.1.2 Preprocessing and Released Representation

a) Unified on-disk representation.: We release every dataset in the same entity-centric Parquet format. One row represents one entity (e.g., a customer, store, transformer, or tweet); its identifier and global targets are scalar columns, whereas time and event features are stored as aligned list-valued columns. Thus, the ith elements of the time, categorical, and numerical arrays describe the same event. The row also stores the sequence length, a list of valid cutoffs (`shifts`), and local-target arrays aligned one-to-one with those cutoffs. Target columns follow the self-describing convention: `target__<name>__<scope>__<metrics>` for example, `target__reg_amount__local__r2`. Train and test outputs are partitioned into Parquet shards, so the released files can be streamed without reconstructing sequences from the original event tables.

[2]https://www.kaggle.com/datasets/mrmorj/alfabattle-20

[3]https://ods.ai/competitions/sberbank-sirius-lesson/data

[4]https://ods.ai/competitions/x5-retailhero-uplift-modeling/data

[5]https://www.kaggle.com/datasets/alexxl/zvuk-dataset

[6]https://huggingface.co/datasets/yandex/yambda

[7]https://remaplab.deib.polimi.it/resources/

[8]https://tianchi.aliyun.com/dataset/46

[9]https://www.timeseriesclassification.com/description.php?Dataset=ElectricDevices

[10]https://github.com/zhouhaoyi/ETDataset

[11]https://www.kaggle.com/competitions/favorita-grocery-sales-forecasting/data

[12]https://www.kaggle.com/competitions/rossmann-store-sales/data

[13]https://www.kaggle.com/datasets/kazanova/sentiment140/data

b) Feature encoding.: Categorical values are replaced with integer frequency ranks: code 1 is assigned to the most frequent value, larger codes to progressively rarer values, and 0 is reserved for missing or unknown categories. We release these mappings together with the processed data. All real-valued input features are normalised. Heavy-tailed quantities such as transaction amounts use a signed $\log(1 + |x|)$ transform; other numerical channels are divided by a robust dataset-specific scale derived from the 0.99 quantile. Time is stored as a relative numerical coordinate rather than an absolute datetime. The reference point follows the source representation: AlphaBattle measures hours from each client's first transaction; datasets supplied with calendar timestamps use offsets from a fixed dataset-level date; and regularly sampled series use the observation index. The resulting time channel is rescaled together with the other numerical features. These transformations place heterogeneous numerical and temporal channels on comparable scales without discarding their within-sequence variation.

c) Shift construction and target storage.: A shift s is a cutoff in an entity's stored arrays: the encoder observes the prefix preceding the target window, and each local target is computed from observations after that cutoff. We retain only cutoffs with at least two preceding observations and enough future context to cover the task horizon in Table S1; entities without a valid cutoff are removed. From the remaining interval we sample up to a dataset-specific maximum number of shifts without replacement and store their sorted indices in the `shifts` column. Local targets are lists in the same order, whereas a global target is stored once as a scalar because it applies to the complete entity sequence. This layout allows different history slices to be generated at loading time without duplicating the underlying event arrays. Table S4 reports the resulting task counts and shift limits.

S.2. PRETRAINING METHOD AND BASELINE DETAILS

This section details the implementations summarised in Table III.

S.2.1 Masked Modelling

MLM trains a bidirectional encoder by hiding part of the input and reconstructing it from the remaining context [2], [4], [21]. In our event-sequence implementation, masking can select full events, spans, or individual categorical and numerical fields. Let M_c and M_r be the masked positions for categorical field c and numerical field r, let $\tilde{x}$ be the corrupted sequence, and let $\hat{x}_i^{(r)}$ be the reconstruction head prediction for numerical field r. The reconstruction terms are

$$\begin{aligned} \mathcal{L}_{\text{MLM-CE}} &= -\sum_{c} \frac{1}{|M_c|} \sum_{i \in M_c} \log p_\theta\left(x_i^{(c)} \mid \tilde{x}\right), \\ \mathcal{L}_{\text{MLM-MSE}} &= \sum_{r} \frac{1}{|M_r|} \sum_{i \in M_r} \left(\hat{x}_i^{(r)} - x_i^{(r)}\right)^2. \end{aligned} \tag{S1}$$

The final MLM loss is a weighted sum of the categorical CE and numerical MSE terms. Masking creates explicit prediction targets: the model must recover the masked content itself.

TABLE S1

CLASSIFICATION AND ANOMALY TARGET DETAILS AND TARGET HORIZONS. CLASS BALANCES AND ANOMALY RATES ARE PERCENTAGES. ROSSMANN AND TWITTER CLASS BALANCES ARE COMPUTED OVER ALL ENTITIES IN THE RELEASED TRAINING REPRESENTATION, BEFORE PARTITIONING BY THE GLOBAL TRAIN/TEST FLAG. "G" DENOTES A GLOBAL TARGET, NUMERIC VALUES GIVE LOCAL-TARGET HORIZONS, "STEP" DENOTES THE NEXT OBSERVATION, AND × MEANS UNAVAILABLE. ROW COLORS DENOTE TRX (GREEN), RECSYS (BLUE), TS (ORANGE), AND TEXT (GRAY).

Dataset	Domain	Classification target	Class balance	Anomaly target	Rate	Horizon			
						C	R	F	A
AGE	TRX	Age group, 4 classes [d]	24.8 : 25.0 : 25.2 : 25.0	Payment-amount coefficient of variation above the 95th percentile	5.00	G	30d	1d	G
AlphaBattle		Application product, 5 product IDs	51.3 : 29.8 : 6.2 : 7.9 : 4.7	Client default [d]	2.83	G	30d	1h	G
Retail		Age quantile, 4 classes [d]	26.8 : 22.1 : 26.9 : 24.1	Amount paid with loyalty points greater than 10	11.15	G	10d	1h	10d
Taobao		Purchase/view ratio, 2 quantiles	75 : 25	Item purchased immediately after being liked	4.59	48h	48h	1h	G
Zvuk		Future listening genre	Top 5: 27.97 : 6.23 : 5.31 : 3.54 : 2.83	Listening-duration coefficient of variation above the 95th percentile	5.00	40d	40d	1d	40d
30Music	RecSys	Listening-time variability, 4 classes	25 : 25 : 25 : 25	Track diversity > 0.95 and mean duration < 0.05	0.24	G	3d	1d	G
Yambda		Recommended vs. organic listening [d]	53.0 : 47.0	Dislike-to-listen ratio above the 95th percentile	5.01	G	10d	1h	G
Electric Devices	TS	Device type [d]	26.98 : 25.02 : 16.50 : 9.54 : 8.14 : 8.13 : 5.69	×	×	G	×	1 step	×
ETT		×	×	×	×	×	×	1 step	×
Favorita		Store type, 5 classes	33.96 : 28.30 : 15.09 : 15.09 : 7.55	Store is the only one in its city	21.00	G	30d	1d	G
Rossmann		Store type [d]	53.99 : 31.21 : 13.27 : 1.52	Max-to-median sales ratio over the next two months	5.00	G	60d	1d	60d
Twitter	Text	Sentiment [d]	50.00 : 50.00	Special-character count normalised by tweet length	2.00	G	G	×	G

This differs from contrastive masking, where masked views are used as augmentations and the masked values are not reconstructed. Our search space varies whether complete events, contiguous spans, or individual categorical and numerical fields are masked, together with the probabilities of random replacement and leaving selected events unchanged. The downstream embedding is taken from the pretrained bidirectional encoder after the reconstruction head is discarded.

S.2.2 Joint-Embedding Predictive Learning

JEPA predicts in representation space rather than reconstructing raw observations [28], [47]. Let C be the visible context positions, x_C the corresponding context events, and T the masked target positions. A context encoder E_θ maps the visible events to context tokens, a target encoder $E_{\bar{\theta}}$ maps the full sequence to target representations, and a predictor P_ϕ maps the context tokens, conditioned on target-position embeddings, into the target representation space. The target branch is stop-gradient and its parameters are not updated by backpropagation through the loss.

$$\mathcal{L}_{\text{JEPA}} = \frac{1}{|T|} \sum_{i \in T} \text{SmoothL1}(p_i, \text{sg}(z_i)), \qquad p_i = P_\phi(E_\theta(x_C), i), \qquad z_i = H_{\bar{\theta}}(x)_i. \tag{S2}$$

Here $\text{sg}(\cdot)$ denotes stop-gradient, $H_{\bar{\theta}}$ denotes the EMA target encoder followed by the target projection head, and p_i is the predictor output for target position i. The target parameters are updated by exponential moving average, $\bar{\theta} \leftarrow \mu\bar{\theta} + (1-\mu)\theta$, after the online update. This is the defining JEPA mechanism for our benchmark: context encoder, EMA target encoder, predictor, and a latent-space target. We tune the EMA coefficient, predictor width, target projection, positional conditioning, and whether downstream embeddings are extracted from the online or target encoder.

S.2.3 Contrastive Learning

Our contrastive methods differ along two axes: how views of a sequence are generated and which contrastive loss is applied. CoLES uses same-entity random slices with the original margin loss. Slice-InfoNCE uses the same type of random slice views but replaces the margin loss with an InfoNCE objective. Mask-InfoNCE keeps the InfoNCE objective and replaces slicing with independently masked views.

S.2.3.1 CoLES [1]: builds positive pairs by sampling two subsequences from the same entity history and negative pairs from different entity histories. Let u and v be two sampled subsequences, $\Phi(\cdot)$ the sequence encoder, $c_u = \Phi(u)$ and $c_v = \Phi(v)$ their embeddings, $d_{uv} = \|c_u - c_v\|_2$, $Y_{uv} = 1$ for a positive same-entity pair and $Y_{uv} = 0$ otherwise, and ρ the margin for negative pairs. CoLES optimises the margin contrastive loss

$$\mathcal{L}_{uv}^{\text{CoLES}} = \frac{1}{2} Y_{uv} d_{uv}^2 + \frac{1}{2}(1 - Y_{uv}) \max(0, \rho - d_{uv})^2. \tag{S3}$$

This objective pulls subsequences from the same entity together and pushes different-entity subsequences apart until the

TABLE S2

PROCESSED SEQUENCE AND LENGTH STATISTICS. "SEQUENCES" IS THE NUMBER OF ROWS AND "EVENTS" IS THE SUM OF `_SEQ_LEN`. VALUES ARE COMPUTED FROM THE TEMPORAL TEST REPRESENTATION, WHOSE ROWS CONTAIN FULL HISTORIES OF ENTITIES WITH A VALID POST-BOUNDARY CUTOFF AND THEREFORE MUST NOT BE ADDED TO TEMPORAL TRAIN COUNTS. ALPHABATTLE COMBINES ITS SOURCE-PROVIDED TRAIN AND TEST PARTITIONS BECAUSE THEY CONTAIN DISTINCT ENTITIES. TWITTER HAS ONLY GLOBAL TARGETS AND NO TEMPORAL TEST REPRESENTATION, SO ITS ROW REPORTS THE TOP-LEVEL TRAINING REPRESENTATION.

Dataset	Sequences	Events	Min	Mean	P90	P99	Max
AGE	29,964	26,420,179	700	881.73	1,071	1,143	1,150
AlphaBattle	1,457,032	443,407,241	6	304.32	734	1,539	9,580
Retail	212,322	33,490,385	6	157.73	299	560	2,513
Taobao	7,437	11,289,272	25	1,517.99	3,358.60	7,506.56	31,030
Zvuk	62,753	143,643,940	10	2,289.04	4,905.80	12,382	132,613
30Music	21,572	22,734,503	11	1,053.89	1,800	8,326.70	89,124
Yambda	8,280	46,355,169	11	5,598.45	13,913.80	24,482.21	27,699
Electric Devices	8,926	856,896	96	96.00	96	96	96
ETT	206	138,432	672	672.00	672	672	672
Favorita	54	90,936	1,684	1,684.00	1,684	1,684	1,684
Rossmann	1,115	1,017,209	758	912.30	942	942	942
Twitter	1,600,000	117,745,496	6	73.59	130	140	373

margin is satisfied. The loss therefore encourages stable entity-level information to be preserved across different temporal windows.

S.2.3.2 InfoNCE-based methods: Following the SimCLR two-view framework [46], the InfoNCE variants produce $2N$ views from a mini-batch of N sequences. Let h_i be the encoder output for view i, z_i the projection-head output used only for the contrastive loss, s_{ik} the cosine similarity between projected views i and k, (i, j) a positive pair of two views of the same sequence, and $\mathcal{A}(i) = \{1, \ldots, 2N\} \setminus \{i\}$ the candidate set for anchor i:

$$s_{ik} = \frac{z_i^\top z_k}{\|z_i\|_2 \|z_k\|_2}. \tag{S4}$$

The InfoNCE loss for anchor i and positive j is

$$\ell_{i,j}^{\text{InfoNCE}} = -\log \frac{\exp(s_{ij}/\tau)}{\exp(s_{ij}/\tau) + \sum_{k \in \mathcal{A}(i) \setminus \{j\}} \exp(s_{ik}/\tau)}. \tag{S5}$$

The numerator is the positive pair; the denominator contains the positive pair and all other views in the batch as negatives. The final loss averages $\ell_{i,j}^{\text{InfoNCE}}$ over positive pairs. Our implementation also supports an angular margin on the positive similarity, but the objective family remains InfoNCE. Mask-InfoNCE is not masked reconstruction: it does not predict masked values, and masking only defines invariances between augmented views. After pretraining, the projection head is discarded and the encoder output is used as the downstream embedding.

S.2.4 Next-Token Prediction

NTP trains a causal model with a next-event prediction loss, as in next-item sequential recommendation models such as SASRec [25]. Let $x = (x_1, \ldots, x_T)$ be an event sequence, let h_t be the hidden state produced from the prefix $x_{\le t}$, and let $x_{t+1}^{(c)}$ and $x_{t+1}^{(r)}$ denote categorical and numerical fields of the next event. The next-event reconstruction terms are

$$\begin{aligned} \mathcal{L}_{\text{NTP-CE}} &= -\sum_{t=1}^{T-1} \sum_{c} \log p_\theta\left(x_{t+1}^{(c)} \mid x_{\le t}\right), \\ \mathcal{L}_{\text{NTP-MSE}} &= \sum_{t=1}^{T-1} \sum_{r} \left(\hat{x}_{t+1}^{(r)} - x_{t+1}^{(r)}\right)^2. \end{aligned} \tag{S6}$$

The final loss is the weighted sum of CE and MSE terms over valid next-event positions. We implement NTP with two backbones: a causal Transformer/GPT-style encoder (NTP-GPT) or a GRU encoder (NTP-GRU). In both cases, the reconstruction head is used only for pretraining and the hidden states are aggregated into downstream embeddings. NTP is used here only as a pretraining objective for embeddings, not as a downstream forecasting task.

S.2.5 Random Encoder Baselines

R-RNN and R-Tr are frozen, randomly initialised GRU and GPT-style Transformer encoders, respectively. They use the same input representation and embedding extraction rules as the corresponding pretrained encoders, isolating the contribution of SSL pretraining from that of the architecture and input encoding.

S.2.6 TSFresh Baseline

TSFresh [9] is included as a classical non-neural baseline. It maps each sequence to a fixed vector of hand-crafted time-series statistics rather than learning a parametric encoder. Our goal is not to tune a separate feature-engineering recipe per dataset, but to provide a strong, comparable classical reference under the same downstream protocol as the SSL methods.

a) Input representation.: For every dataset we use the same fixed-order preprocessed splits as the neural methods. Each entity is identified by its native index column

TABLE S3
CATEGORICAL AND NUMERICAL INPUTS IN THE RELEASED REPRESENTATION. CATEGORICAL ENTRIES REPORT THE ORIGINAL CARDINALITY; ARROWS DENOTE CARDINALITY CAPS APPLIED BY THE MODEL. "NOT USED" MARKS STORED FEATURES EXCLUDED FROM MODEL INPUTS, AND "–" DENOTES AN UNAVAILABLE INPUT TYPE.

Dataset	Categorical features and cardinalities	Numerical model inputs
AGE	small_group: 202	amount_rur
AlphaBattle	card_type: 173; city: 163; country: 24; currency: 11; day_of_week: 7; ecommerce_flag: 3; income_flag: 3; mcc: 108; mcc_category: 28; operation_kind: 7; operation_type: 22; operation_type_group: 4; payment_system: 7; weekofyear: 53 (not used)	hour; amnt; hour_diff
Retail	is_alcohol: 2; is_own_trademark: 2; level_1: 3; level_2: 42; level_3: 201; level_4: 788; product_id: 42,522 (not used); segment_id: 116	trn_sum_from_iss; netto; regular_points_received; express_points_received; product_quantity; trn_sum_from_red; regular_points_spent; express_points_spent; purchase_sum
Taobao	behavior_type: 4; item_category: 8,916	–
Zvuk	cluster_id: 128; track_id: 1,506,950 → 20,000	play_duration
30Music	track_id: 4,519,104 → 15,000	play_duration
Yambda	event_type: 5; is_organic: 2; item_id: 934,057 → 15,000	played_ratio_pct; track_length_seconds
Electric Devices	–	sequence
ETT	–	HUFL; HULL; MUFL; MULL; LUFL; LULL
Favorita	class_id: 334 (not used)	class_1_sales–class_334_sales
Rossmann	DayOfWeek: 7; Open: 2; Promo: 2; SchoolHoliday: 2; StateHoliday: 1	Sales; Customers
Twitter	char: 190	–

(client_id, tweet_id, Store, and so on). From the event-time axis we compute first-order inter-arrival gaps

$$\Delta t_i = t_i - t_{i-1}, \tag{S7}$$

with $\Delta t_1 = 0$. For text-like sequences without timestamps, the same construction is applied to character-position indices. This yields one univariate series per entity (global mode) or one series per temporal cutoff (local mode).

b) Feature extraction.: Features are computed with the library's EfficientFCParameters set. This configuration evaluates a broad catalogue of time-, frequency-, and signal-processing statistics while omitting the most expensive entropy-based calculators. For a single time_delta channel the resulting representation has fixed dimensionality 777; we keep *all* of these features.

c) TSFresh Feature-Importance Analysis: Although the benchmark uses the full 777-dimensional TSFresh vector, we additionally study *which* statistical families carry predictive signal. This analysis is separate from embedding construction: no features are removed before downstream evaluation.

For each dataset and downstream target, we fit an auxiliary gradient-boosted regressor (XGBRegressor) on the *training* TSFresh embeddings and record the top-25 features by importance. Targets are processed separately for global and local settings; for local targets, labels are aligned to the corresponding shift index.

We then aggregate these rankings across datasets and task families to identify recurrent feature types. In particular, we track how often FFT- and quantiles appear among the top predictors, and compare variants with and without explicit sequence-length statistics.

Figure S1 is constructed from one-hot encodings of the top TSFresh feature families per dataset–task pair. Hierarchical clustering of these profiles yields the dendrogram on the left; the stacked bars on the right show the corresponding family shares within each pair (normalized to sum to 1).

Five clusters emerge, and the same grouping is visible in both the dendrogram and the bar profiles. The first two clusters are dominated by regression-type targets (regression and forecasting). The next two are dominated by classification-type targets (classification and anomaly detection). The remaining cluster is more heterogeneous and sits farthest from the others in the hierarchy, suggesting a residual group whose top-feature composition differs most strongly from the rest.

S.3. METHOD HYPERPARAMETERS

a) Training and stopping.: Each HPO trial trains the encoder only on its SSL objective. CoLES, Slice-InfoNCE, Mask-InfoNCE, NTP-GRU, and NTP-GPT use early stopping on the SSL validation loss, with dataset-specific patience between three and ten epochs. MLM and JEPA instead run for a fixed 100 epochs. After the run terminates, the encoder is frozen and its embeddings are evaluated once on every available downstream task; these metrics form the multi-target Optuna objective. The downstream validators therefore select among completed runs but do not determine their training trajectories.

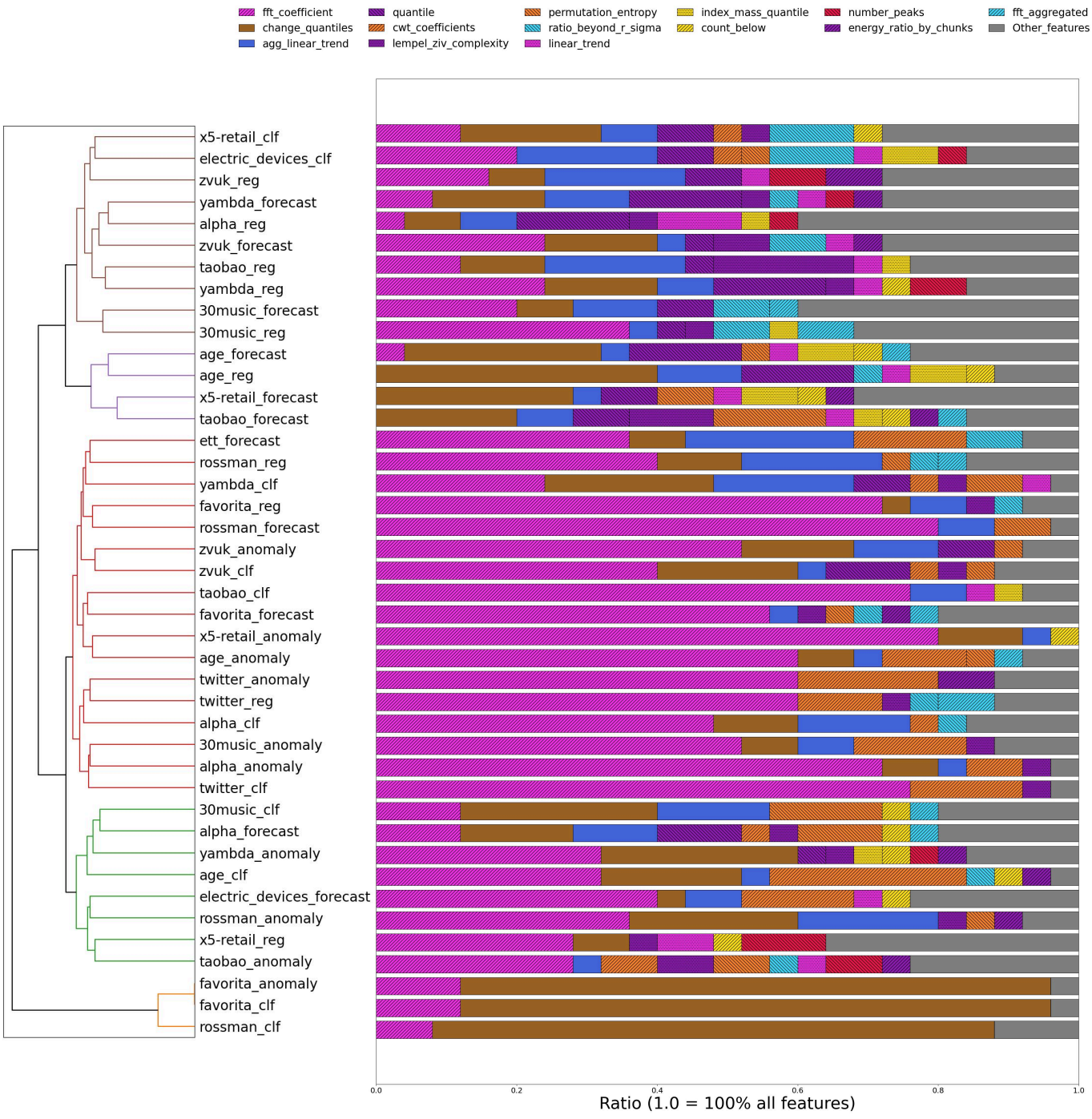


Fig. S1. TSFresh feature-family analysis across dataset–task pairs. **Left:** hierarchical clustering of top-feature profiles. **Right:** normalized family shares within the top-25 XGB importances per pair (shares sum to 1 within each row). FFT- and quantile-related families recur across domains and tasks, but their relative weight varies by dataset and downstream objective.

TABLE S4

DOWNSTREAM SAMPLE CONSTRUCTION IN THE RELEASED REPRESENTATION. SHIFT COUNTS ARE PER-ENTITY MAXIMA; SHORTER HISTORIES MAY CONTRIBUTE FEWER LOCAL EXAMPLES. GLOBAL SPLIT COUNTS ARE OBTAINED FROM THE `GLOBAL_TRAIN` FLAG IN THE TOP-LEVEL TRAINING PARQUET AND IMPLEMENT THE APPROXIMATELY 90/10 ENTITY SPLIT. FOR ALPHABATTLE, THESE GLOBAL COUNTS ARE COMPUTED WITHIN THE SOURCE TRAINING PARTITION, WHEREAS ITS LOCAL TEST HISTORIES COME FROM THE SOURCE TEST PARTITION. DASHES DENOTE DATASETS WITHOUT A TARGET OF THE CORRESPONDING SCOPE.

Dataset	Local tasks	Global tasks	Max shifts/entity		Global split	
			Train	Test	Train	Test
AGE	2	2	9	1	27,000	3,000
AlphaBattle	2	2	9	1	861,481	95,721
Retail	3	1	9	1	318,074	35,342
Taobao	3	1	9	1	8,749	973
Zvuk	4	0	9	1	–	–
30Music	2	2	9	1	35,934	3,993
Yambda	2	2	9	1	8,277	920
Electric Devices	1	1	27	3	8,033	893
ETT	1	0	90	10	–	–
Favorita	2	2	90	10	48	6
Rossmann	3	1	90	10	1,003	112
Twitter	0	3	0	–	1,440,000	160,000

b) Compute budget.: MLM and JEPA have larger conditional search spaces and use a time-bounded study: optimisation terminates at the study-level plateau criterion or after three days, resulting in approximately 150–300 trials depending on the dataset. Figure S2 provides a post-hoc convergence analysis for the fixed-budget NTP studies.

All neural methods tune learning rate η, weight decay λ, backbone width h, backbone depth L, categorical and numerical feature-embedding dimensions d_{cat} and d_{num}, and hidden-state pooling (mean or last). Table S5 gives their method-specific ranges together with the additional architectural, augmentation, and objective parameters. The subscript "log" denotes logarithmic sampling. The budget column reports the maximum number of trials or wall-clock limit, with the number of initial random trials in parentheses.

S.4. FULL RESULTS TABLES

S.4.1 Per-Task Results

Tables S7–S10 report the per-dataset results underlying Table I, one table per task family. Cells report the mean downstream metric and its standard deviation across repeated runs; the two random encoders (R-RNN, R-Tr) are shown in separated reference columns and do not participate in the eight-method ranks printed in these absolute-score tables. Table I and the rank-profile and rank-bar figures instead include all ten methods. Superscripts show the positions occupied by each uncertainty-aware rank group over the eight non-random methods. Groups spanning two positions list both positions (e.g., $1, 2$), whereas groups spanning three or more positions are written as a range (e.g., 1-4); each method receives the corresponding mid-rank when results are aggregated. Columns are ordered by the mean rank over the table's datasets, reported in the Average rank row; per row, the best group is in bold and the second best is underlined. Row colors follow the domain grouping of Table II. Degenerate settings are directly visible: on Favorita regression every pretrained encoder scores $R^2 < 0$ while the RNN random encoder does not, and on ETT forecasting only Slice-InfoNCE and Mask-InfoNCE improve over a constant predictor.

For completeness, Table S6 provides the complementary dataset-centred view. It ranks all ten methods within each dataset-task setting and then averages over the tasks available for each dataset.

The comparison with random encoders is shown in Figure 5 in Section VI-A; the tables below provide the corresponding absolute scores.

Figure S3 complements the tables with the same uncertainty-aware ranks, aggregated overall and by task family. Each panel shows the mean rank of every method with one standard deviation across its datasets-task settings; methods are sorted within each panel. The overall panel matches the column averages of Table S6 up to weighting: here every dataset-task pair contributes equally, whereas Table S6 first averages within a dataset. The large standard deviations, often exceeding two rank positions, show that no method is consistently placed across datasets, and that single-dataset comparisons are unreliable for method selection.

S.4.2 Aggregation Selection Details

Figures S4 and S5 detail the aggregation choices summarised in Section VI-B.

At the method level (Figure S4), the methods included in this analysis follow the task-driven pattern, but with different strength. CoLES is a notable outlier: it is the only method that prefers the last hidden state even on globally targeted tasks (59% on global targets vs. 24–41% for the other methods), selecting it in 70% of anomaly and 64% of classification configurations. A plausible explanation is that CoLES is trained to make embeddings of random subsequences of the same

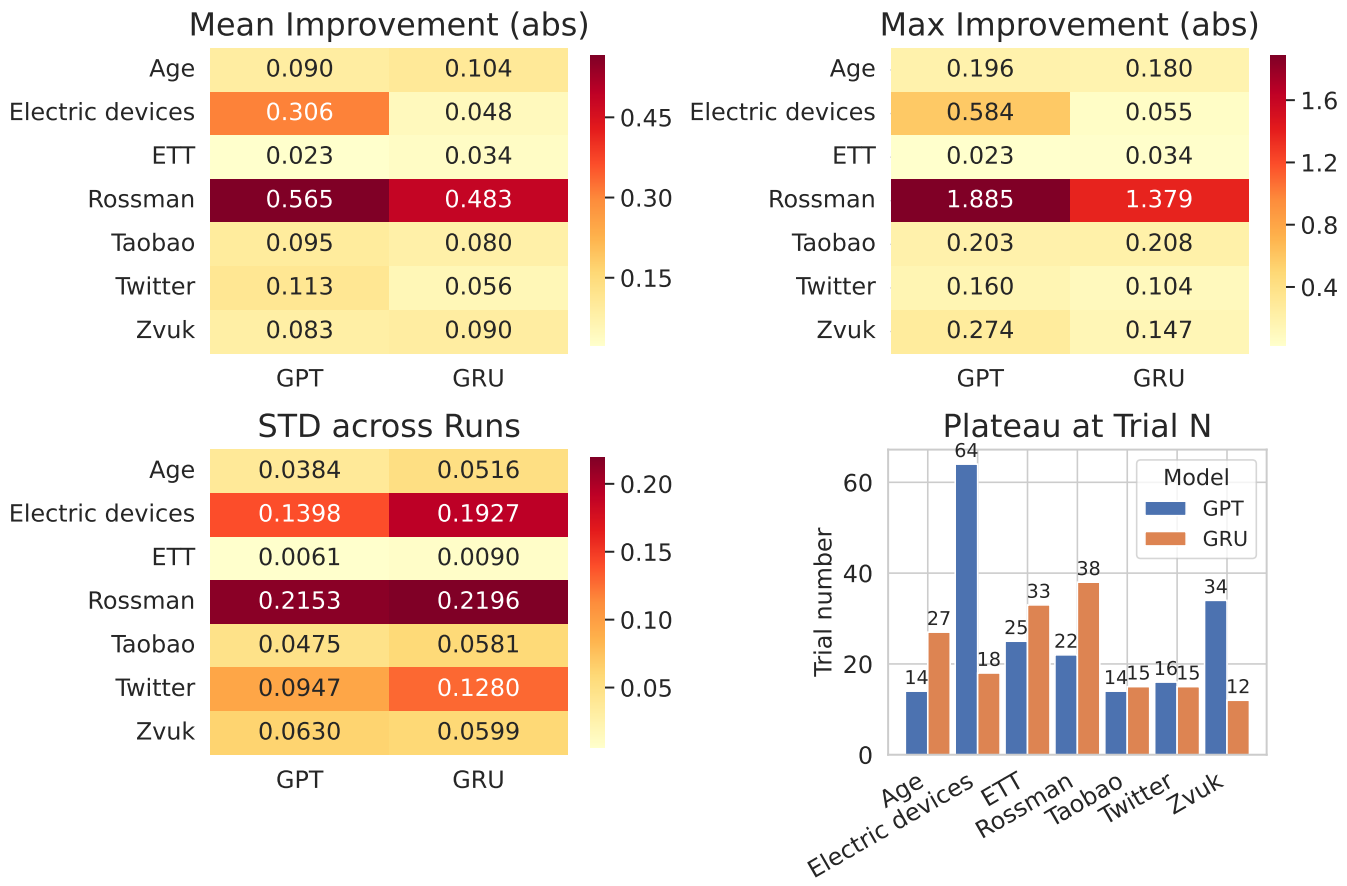


Fig. S2. Post-hoc convergence analysis for the fixed-budget NTP studies. The heatmaps summarise the absolute downstream-metric improvement obtained during HPO and its variation across repeated studies; the lower-right panel reports the first trial whose best value is within 2% of the final study best. This diagnostic motivates the 70-trial NTP budget and is not itself an early-stopping rule.

user agree, so its recurrent state at any position, including the final one, is already optimised to be a sequence-level summary, and mean-pooling adds little. In contrast, NTP-GPT is the most mean-leaning method on global targets (18% last on classification, 20% on anomaly): its per-step states are optimised for next-event prediction, so entity-level information has to be recovered by averaging. Forecasting is unanimous: every analysed method selects the last state in at least 70% of configurations, and NTP-GRU in all of them.

At the dataset level (Figure S5), the choices track the semantics of individual targets. Twitter selects mean-pooling in every configuration across all tasks and methods included in this analysis, consistent with its lack of an event-time axis: all its targets are global sequence-level properties. Favorita anomaly detection is likewise unanimous for mean-pooling; its anomaly target (whether a store is the only one in its city) is a static property of the entity rather than a temporal event. At the other extreme, AGE regression selects the last state for every analysed method, matching its short-horizon local target (purchase amount over the next days), and Taobao is the most recency-driven dataset overall (79% last across tasks). The overall spread across datasets, from 0% to 79% last, shows that the aggregation choice reflects properties of the target definition rather than noise in hyperparameter search.

S.4.3 Validator Sensitivity

The main text reports the final ten Optuna trials, which reflect configurations reached near the end of optimisation. Here the first ten, approximately random trials provide a control for the initial exploration stage. Table S11 compares pairwise validator correlations between these stages, and Table S12 reports the corresponding ranks and regret.

S.4.4 Tucker-Based Feature Fusion

a) Data Representation.: Assume that each sample is represented by two feature vectors (views)

$$\mathbf{x}^{(n)} \in \mathbb{R}^{F_1}, \qquad \mathbf{y}^{(n)} \in \mathbb{R}^{F_2}, \tag{S8}$$

where $n = 1, \ldots, N$. The training dataset is

$$\mathcal{D} = \left\{ \left(\mathbf{x}^{(n)}, \mathbf{y}^{(n)} \right) \right\}_{n=1}^{N}. \tag{S9}$$

The objective is to learn compact representations that preserve the interactions between the two feature spaces.

TABLE S5

HYPERPARAMETER SEARCH SPACES. THE MIDDLE COLUMN CONTAINS PARAMETER GROUPS SHARED BY ALL NEURAL METHODS; THE NEXT COLUMN CONTAINS ONLY METHOD-SPECIFIC PARAMETERS AND FIXED DESIGN CHOICES.

Method	Common search parameters	Additional parameters and fixed design	HPO budget
CoLES	$\eta : [10^{-5}, 10^{-2}]_{\log}$; $\lambda : [10^{-10}, 10^{-2}]_{\log}$; $h : [64, 1024]_{\log}$; L : 1–5; $d_{\text{cat}}, d_{\text{num}}$: 1–128; pool $\in \{\text{mean}, \text{last}\}$	**Tuned:** numerical normalisation on/off. **Fixed:** five slices, five hard negatives, margin $\rho = 1$.	100 (7 random)
Slice-InfoNCE	$\eta : [10^{-5}, 10^{-2}]_{\log}$; $\lambda : [10^{-10}, 10^{-2}]_{\log}$; $h : [64, 1024]_{\log}$; L : 1–5; $d_{\text{cat}}, d_{\text{num}}$: 1–128; pool $\in \{\text{mean}, \text{last}\}$	**Tuned:** temperature $\tau : [10^{-2}, 10^{2}]_{\log}$; time encoding $\in \{\text{event}, \Delta t\}$; numerical normalisation on/off. **Fixed:** two independently sampled slices.	100 (7 random)
Mask-InfoNCE	$\eta : [10^{-5}, 10^{-2}]_{\log}$; $\lambda : [10^{-10}, 10^{-2}]_{\log}$; $h : [64, 1024]_{\log}$; L : 1–5; $d_{\text{cat}}, d_{\text{num}}$: 1–128; pool $\in \{\text{mean}, \text{last}\}$	**Tuned:** temperature $\tau : [10^{-2}, 10^{2}]_{\log}$; time encoding $\in \{\text{event}, \Delta t\}$; numerical normalisation on/off. **Fixed:** two independently masked views.	100 (7 random)
MLM	$\eta : [10^{-5}, 10^{-2}]_{\log}$; $\lambda : [10^{-10}, 10^{-2}]_{\log}$; $h : [64, 1024]_{\log}$; L : 1–5; $d_{\text{cat}}, d_{\text{num}}$: 16–128 (step 16); pool $\in \{\text{mean}, \text{last}\}$	**Architecture:** heads $\in \{1, 2, 4, 8\}$; block passes $\in \{1, 2\}$; dropout $[0, 0.3]$; output size $\in \{64, 256, 1024\}$. **Masking:** probability $[0.1, 0.8]$; span length 5–100; span share $[0.1, 0.9]$; random replacement $[0, 0.2]$; unchanged event $[0, 0.15]$; partial categorical/numerical masking $[0, 0.5]$.	≤ 3 days (10 random)
JEPA	$\eta : [10^{-5}, 10^{-2}]_{\log}$; $\lambda : [10^{-10}, 10^{-2}]_{\log}$; $h : [128, 1024]_{\log}$; L : 1–4; $d_{\text{cat}}, d_{\text{num}}$: 32–128 (step 16); pool $\in \{\text{mean}, \text{last}\}$	**Architecture:** heads $\in \{1, 2, 4, 8\}$; block passes $\in \{1, 2\}$; dropout $[0, 0.2]$; output size $\in \{64, 256, 1024\}$. **JEPA:** EMA $\in \{0.99, 0.999, 0.9999\}$; downstream branch $\in \{\text{online}, \text{target}\}$; predictor multiplier $\in \{0.5, 1, 2\}$; target projection $\in \{\text{predictor output}, \text{online target}\}$; mask probability $[0.1, 0.8]$.	≤ 3 days (10 random)
NTP-GRU	$\eta : [10^{-5}, 10^{-2}]_{\log}$; $\lambda : [10^{-10}, 10^{-2}]_{\log}$; $h : [64, 1024]_{\log}$; L : 1–5; $d_{\text{cat}}, d_{\text{num}}$: 8–128 (step 8); pool $\in \{\text{mean}, \text{last}\}$	Numerical normalisation on/off; output size $\in \{64, 256, 1024\}$; GRU backbone.	70 (7 random)
NTP-GPT	$\eta : [10^{-5}, 10^{-2}]_{\log}$; $\lambda : [10^{-10}, 10^{-2}]_{\log}$; $h : [64, 1024]_{\log}$; L : 1–5; $d_{\text{cat}}, d_{\text{num}}$: 8–128 (step 8); pool $\in \{\text{mean}, \text{last}\}$	Numerical normalisation on/off; output size $\in \{64, 256, 1024\}$; heads $\in \{1, 2, 4, 8\}$; GPT-style causal Transformer backbone.	70 (7 random)

b) Feature Interaction Model.: We assume that every feature of the first view may interact with every feature of the second view. The interaction is modeled by the outer product

$$\mathbf{X}^{(n)} = \mathbf{x}^{(n)} \otimes \mathbf{y}^{(n)}, \tag{S10}$$

whose (i, j)-th entry is

$$X_{ij}^{(n)} = x_i^{(n)} y_j^{(n)}. \tag{S11}$$

Thus each sample is represented by a rank-one fusion matrix

$$\mathbf{X}^{(n)} \in \mathbb{R}^{F_1 \times F_2}. \tag{S12}$$

Collecting all samples forms the third-order fusion tensor

$$\mathcal{X} \in \mathbb{R}^{F_1 \times F_2 \times N}, \tag{S13}$$

whose n-th frontal slice is

$$\mathcal{X}(:, :, n) = \mathbf{X}^{(n)} = \mathbf{x}^{(n)} \mathbf{y}^{(n)T}. \tag{S14}$$

c) Problem Formulation.: Since the fusion tensor can be extremely large, our objective is to find two projection matrices

$$\mathbf{B} \in \mathbb{R}^{F_1 \times r_1}, \qquad \mathbf{C} \in \mathbb{R}^{F_2 \times r_2}, \tag{S15}$$

where $r_1 \ll F_1$, $r_2 \ll F_2$, such that the interaction information contained in $\mathcal{X}$ is preserved as much as possible. The projections are obtained by computing a low-rank Tucker approximation

$$\mathcal{X} \approx \mathcal{G} \times_1 \mathbf{B} \times_2 \mathbf{C}, \tag{S16}$$

where $\mathcal{G} \in \mathbb{R}^{r_1 \times r_2 \times N}$ is the Tucker core tensor.

The corresponding optimization problem is

$$\begin{aligned} \min_{\mathbf{B}, \mathbf{C}, \mathcal{G}} \quad & \left\| \mathcal{X} - \mathcal{G} \times_1 \mathbf{B} \times_2 \mathbf{C} \right\|_F^2, \\ \text{s.t.} \quad & \mathbf{B}^T \mathbf{B} = \mathbf{I}, \\ & \mathbf{C}^T \mathbf{C} = \mathbf{I}. \end{aligned} \tag{S17}$$

d) Equivalent Optimization.: For fixed projection matrices, the optimal Tucker core is

$$\mathcal{G} = \mathcal{X} \times_1 \mathbf{B}^T \times_2 \mathbf{C}^T. \tag{S18}$$

TABLE S6

AVERAGE UNCERTAINTY-AWARE RANK OF EACH METHOD PER DATASET (LOWER IS BETTER), AVERAGED OVER THE DOWNSTREAM TASKS AVAILABLE FOR THAT DATASET. METHODS ARE ORDERED BY THEIR COLUMN AVERAGE; THE BEST VALUE PER ROW IS IN **BOLD**, THE SECOND BEST IS UNDERLINED. ROW COLORS GROUP DATASETS BY DOMAIN AS IN TABLE II: TRANSACTIONS (GREEN), RECOMMENDER SYSTEMS (BLUE), TIME SERIES (ORANGE), TEXT (GRAY).

Dataset	CoLES	Mask-InfoNCE	Slice-InfoNCE	NTP-GRU	NTP-GPT	JEPA	MLM	TSFresh	R-Tr	R-RNN
AGE	**2.75**	4.00	3.25	3.12	3.88	5.50	6.50	9.50	8.75	7.75
AlphaBattle	**2.25**	4.00	5.75	3.50	4.12	4.75	7.25	6.38	8.38	8.62
Retail	2.00	**1.75**	2.75	4.12	7.38	5.38	7.00	8.12	7.75	8.75
Taobao	**2.00**	3.88	2.38	5.25	7.50	4.00	5.62	8.75	6.50	9.12
Zvuk	**1.75**	5.88	1.88	6.12	4.38	8.38	5.88	6.00	5.12	9.62
30Music	3.00	**1.75**	6.75	6.00	6.12	3.62	6.75	4.50	7.75	8.75
Yambda	4.50	2.88	**2.62**	4.62	4.50	6.00	6.88	5.38	7.75	9.88
Rossmann	2.50	**2.00**	4.50	5.88	4.88	3.88	4.88	9.00	8.88	8.62
ETT	6.50	**1.50**	**1.50**	6.50	3.50	3.50	6.50	9.00	10.00	6.50
Electric Devices	3.25	**2.00**	3.75	2.75	4.00	8.25	6.50	7.25	9.00	8.25
Favorita	4.88	8.12	6.62	3.38	3.50	**3.25**	6.38	6.38	6.00	6.50
Twitter	3.67	4.67	**2.17**	2.33	2.50	7.00	6.00	9.00	9.33	8.33
Average	**3.25**	3.53	3.66	4.47	4.69	5.29	6.34	7.44	7.93	8.39

TABLE S7

REGRESSION: ABSOLUTE R^2 OF EACH METHOD PER DATASET (HIGHER IS BETTER). CONVENTIONS ARE DESCRIBED IN SECTION S.4.1.

Regression	R-RNN	R-Tr	CoLES	Slice-InfoNCE	Mask-InfoNCE	NTP-GRU	TSFresh	NTP-GPT	JEPA	MLM
AGE	0.306 ±0.001	0.313 ±0.002	**0.512** ±0.001^{1}	0.506 ±0.001^{2}	0.498 ±0.001^{3}	0.490 ±0.001^{4}	0.223^{8}	0.440 ±0.001^{5}	0.401 ±0.004^{6,7}	0.405 ±0.002^{6,7}
AlphaBattle	0.393 ±0.001	0.596 ±0.005	**0.713** ±0.002^{1}	0.699 ±0.000^{4}	0.705 ±0.001^{3}	0.710 ±0.000^{2}	0.697^{5}	0.667 ±0.001^{7,8}	0.672 ±0.002^{6}	0.665 ±0.002^{7,8}
Retail	0.041 ±0.000	0.132 ±0.001	**0.159** ±0.004$^{1–3}$	**0.159** ±0.001$^{1–3}$	**0.166** ±0.006$^{1–3}$	0.138 ±0.002^{4}	0.129^{5}	0.105 ±0.004^{6}	0.062 ±0.009^{8}	0.089 ±0.005^{7}
Taobao	-0.021 ±0.005	0.205 ±0.005	**0.254** ±0.008^{1,2}	**0.257** ±0.002^{1,2}	0.168 ±0.001^{5}	0.191 ±0.002^{3,4}	-0.201^{8}	0.074 ±0.003^{7}	0.179 ±0.012^{3,4}	0.103 ±0.023^{6}
Zvuk	-0.016 ±0.000	0.281 ±0.001	**0.252** ±0.010^{1,2}	**0.266** ±0.005^{1,2}	0.052 ±0.006^{7}	0.131 ±0.001^{6}	0.189^{3}	0.172 ±0.003^{4,5}	0.036 ±0.002^{8}	0.162 ±0.007^{4,5}
30Music	0.002 ±0.001	0.113 ±0.003	0.152 ±0.001^{3}	0.006 ±0.002^{7}	0.198 ±0.001^{2}	0.051 ±0.002^{5}	**0.263**^{1}	0.034 ±0.003^{6}	0.117 ±0.011^{4}	-0.020 ±0.022^{8}
Yambda	0.580 ±0.001	0.727 ±0.002	0.749 ±0.001^{3}	**0.763** ±0.001^{1}	0.760 ±0.002^{2}	0.686 ±0.005^{6}	0.745^{4}	0.699 ±0.005^{5}	0.674 ±0.001^{7}	0.611 ±0.003^{8}
Rossmann	0.915 ±0.001	-0.084 ±0.049	**0.971** ±0.001^{1,2}	0.951 ±0.000^{4}	**0.969** ±0.002^{1,2}	0.936 ±0.001^{6}	-0.031^{8}	0.922 ±0.002^{7}	0.947 ±0.000^{5}	0.954 ±0.000^{3}
Favorita	0.163 ±0.043	0.435 ±0.054	-0.742 ±0.129^{3,4}	-1.421 ±0.000^{5,6}	-3.061 ±0.290^{7,8}	-0.154 ±0.017^{2}	**-0.016**^{1}	-0.781 ±0.054^{3,4}	-1.447 ±0.503^{5,6}	-2.393 ±0.805^{7,8}
Twitter	0.280 ±0.002	0.067 ±0.000	0.545 ±0.000^{5}	0.598 ±0.000^{4}	0.609 ±0.000^{3}	**0.689** ±0.000^{1}	-0.017^{8}	0.659 ±0.000^{2}	0.480 ±0.000^{6}	0.478 ±0.000^{7}
Avg. rank	–	–	2.3	3.2	3.6	4.0	5.1	5.3	6.0	6.5

Substituting the optimal core into (S17) yields the equivalent maximization problem

$$\begin{aligned} \max_{\mathbf{B},\mathbf{C}} \quad & \left\| \mathcal{X} \times_1 \mathbf{B}^T \times_2 \mathbf{C}^T \right\|_F^2, \\ \text{s.t.} \quad & \mathbf{B}^T\mathbf{B} = \mathbf{I}, \\ & \mathbf{C}^T\mathbf{C} = \mathbf{I}. \end{aligned} \tag{S19}$$

This formulation seeks the pair of orthonormal projection matrices that maximizes the energy preserved in the projected fusion tensor.

e) Efficient Solution.: We compute the mode-wise HOSVD projections from Gram matrices without explicitly constructing the fusion tensor.

The mode-1 Gram matrix is

$$\begin{aligned} \mathbf{G}_1 &= \mathbf{X}_{(1)}\mathbf{X}_{(1)}^T \\ &= \sum_{n=1}^{N} \mathbf{x}^{(n)}\mathbf{y}^{(n)T}\mathbf{y}^{(n)}\mathbf{x}^{(n)T} \\ &= \sum_{n=1}^{N} \left\|\mathbf{y}^{(n)}\right\|^2 \mathbf{x}^{(n)}\mathbf{x}^{(n)T}. \end{aligned} \tag{S20}$$

Similarly,

$$\mathbf{G}_2 = \sum_{n=1}^{N} \left\|\mathbf{x}^{(n)}\right\|^2 \mathbf{y}^{(n)}\mathbf{y}^{(n)T}. \tag{S21}$$

The projection matrices are obtained from the dominant eigenvectors

$$\mathbf{B} = \mathrm{eig}_{r_1}(\mathbf{G}_1), \qquad \mathbf{C} = \mathrm{eig}_{r_2}(\mathbf{G}_2), \tag{S22}$$

where $\mathrm{eig}_r(\cdot)$ denotes the eigenvectors corresponding to the r largest eigenvalues.

These eigenvectors give the same mode-wise projection subspaces as the truncated HOSVD. They provide a tractable approximation to the joint Tucker optimisation problem [44], [57].

f) Final fused representation.: The projection matrices are fitted on the training embeddings and then kept fixed. TuckerFactorConcat concatenates the two projected views to produce the downstream representation:

$$\mathbf{z} = \begin{bmatrix} \mathbf{B}^T\mathbf{x} \\ \mathbf{C}^T\mathbf{y} \end{bmatrix} \in \mathbb{R}^{r_1+r_2}. \tag{S23}$$

The outer-product construction determines the projection directions; the concatenated projected views form the embedding supplied to the downstream validator.

TABLE S8

CLASSIFICATION: ABSOLUTE ACCURACY OF EACH METHOD PER DATASET (HIGHER IS BETTER). CONVENTIONS ARE DESCRIBED IN SECTION S.4.1.

Classification	R-RNN	R-Tr	Slice-InfoNCE	CoLES	Mask-InfoNCE	NTP-GPT	NTP-GRU	JEPA	MLM	TSFresh
AGE	0.399 ±0.003	0.420 ±0.010	**0.630** ±0.004[1]	0.617 ±0.002[4]	0.598 ±0.001[6]	0.620 ±0.003[2,3]	0.621 ±0.001[2,3]	0.558 ±0.002[7]	0.601 ±0.002[5]	0.418[8]
AlphaBattle	0.517 ±0.002	0.513 ±0.005	**0.523** ±0.009[1–6]	**0.524** ±0.010[1–6]	**0.526** ±0.006[1–6]	**0.518** ±0.003[1–6]	**0.517** ±0.004[1–6]	**0.517** ±0.004[1–6]	0.510 ±0.005[7,8]	0.512[7,8]
Retail	0.372 ±0.006	0.384 ±0.007	**0.516** ±0.004[1,2]	**0.513** ±0.004[1,2]	0.495 ±0.006[3]	0.458 ±0.001[6]	0.463 ±0.001[5]	0.479 ±0.005[4]	0.421 ±0.005[7]	0.315[8]
Taobao	0.663 ±0.000	0.663 ±0.000	0.676 ±0.000[2]	**0.676** ±0.000[1]	0.676 ±0.000[3–5]	0.654 ±0.017[6]	0.676 ±0.000[3–5]	0.676 ±0.000[3–5]	0.562 ±0.034[7]	0.378[8]
Zvuk	0.278 ±0.000	0.377 ±0.002	**0.503** ±0.001[1,2]	**0.502** ±0.001[1,2]	0.489 ±0.003[3]	0.479 ±0.002[4]	0.476 ±0.000[5]	0.339 ±0.000[8]	0.438 ±0.002[6]	0.364[7]
30Music	0.290 ±0.006	0.359 ±0.005	0.683 ±0.006[4]	0.765 ±0.005[3]	0.788 ±0.002[2]	0.449 ±0.001[6]	0.392 ±0.004[7]	**0.872** ±0.004[1]	0.454 ±0.003[5]	0.386[8]
Yambda	0.572 ±0.015	0.581 ±0.014	0.742 ±0.006[5,6]	0.761 ±0.004[4]	0.737 ±0.005[5,6]	**0.814** ±0.006[1]	0.790 ±0.015[2,3]	0.711 ±0.007[7]	0.795 ±0.004[2,3]	0.696[8]
Rossmann	0.560 ±0.005	0.562 ±0.009	**0.729** ±0.022[1–3]	**0.708** ±0.005[1–3]	**0.717** ±0.031[1–3]	0.601 ±0.027[4–7]	0.568 ±0.010[4–7]	0.568 ±0.010[4–7]	0.568 ±0.010[4–7]	0.527[8]
Electric Devices	0.329 ±0.029	0.323 ±0.016	0.833 ±0.002[2,3]	0.832 ±0.005[2,3]	**0.845** ±0.005[1]	0.751 ±0.006[6]	0.807 ±0.011[4,5]	0.603 ±0.014[8]	0.683 ±0.001[7]	0.807[4,5]
Favorita	0.167 ±0.167	0.167 ±0.289	0.500 ±0.000[4,5]	0.444 ±0.255[4,5]	0.389 ±0.096[6,7]	**0.667** ±0.289[1–3]	**0.667** ±0.167[1–3]	**0.611** ±0.096[1–3]	0.389 ±0.096[6,7]	0.000[8]
Twitter	0.564 ±0.004	0.551 ±0.010	**0.752** ±0.000[1,2]	0.747 ±0.000[4]	0.690 ±0.001[5]	**0.752** ±0.001[1,2]	0.751 ±0.000[3]	0.634 ±0.000[8]	0.662 ±0.001[6]	0.636[7]
Avg. rank	–	–	2.7	2.9	3.8	4.0	4.0	5.3	5.9	7.5

TABLE S9

FORECASTING: ABSOLUTE R^2 OF EACH METHOD PER DATASET (HIGHER IS BETTER). CONVENTIONS ARE DESCRIBED IN SECTION S.4.1.

Forecasting	R-RNN	R-Tr	Mask-InfoNCE	NTP-GPT	CoLES	NTP-GRU	JEPA	Slice-InfoNCE	TSFresh	MLM
AGE	0.373 ±0.001	0.292 ±0.001	0.354 ±0.002[4]	0.390 ±0.004[2]	0.385 ±0.001[3]	**0.414** ±0.001[1]	0.330 ±0.007[6,7]	0.344 ±0.001[5]	-0.249[8]	0.327 ±0.001[6,7]
AlphaBattle	-0.006 ±0.003	-0.004 ±0.003	0.025 ±0.004[5–7]	0.050 ±0.000[2]	0.042 ±0.000[4]	**0.057** ±0.001[1]	0.025 ±0.006[5–7]	0.028 ±0.000[5–7]	0.044[3]	0.023 ±0.003[8]
Retail	0.342 ±0.003	0.260 ±0.008	**0.700** ±0.001[1]	0.298 ±0.002[6,7]	0.460 ±0.000[2]	0.412 ±0.002[3,4]	0.403 ±0.108[3,4]	0.359 ±0.002[5]	0.081[8]	0.310 ±0.029[6,7]
Taobao	-0.004 ±0.002	0.033 ±0.003	**0.144** ±0.004[1,2]	0.052 ±0.008[7,8]	0.116 ±0.000[3]	0.049 ±0.000[7,8]	**0.134** ±0.007[1,2]	0.087 ±0.002[5]	0.074[6]	0.107 ±0.006[4]
Zvuk	-0.135 ±0.011	0.040 ±0.002	-0.122 ±0.003[8]	0.016 ±0.001[2,3]	**0.107** ±0.002[1]	-0.097 ±0.001[6,7]	-0.098 ±0.127[6,7]	0.031 ±0.017[2,3]	0.012[4]	-0.001 ±0.003[5]
30Music	-0.005 ±0.000	-0.032 ±0.006	0.124 ±0.004[2]	-0.003 ±0.004[5]	0.085 ±0.002[3]	0.024 ±0.004[4]	-0.019 ±0.002[6]	-0.054 ±0.002[8]	**0.142**[1]	-0.033 ±0.007[7]
Yambda	0.081 ±0.001	0.103 ±0.003	0.175 ±0.007[2,3]	0.149 ±0.002[4–6]	0.135 ±0.004[7]	0.148 ±0.004[4–6]	0.144 ±0.005[4–6]	0.177 ±0.002[2,3]	**0.188**[1]	0.085 ±0.002[8]
Rossmann	-1.114 ±0.000	-0.500 ±0.000	0.106 ±0.000[3]	0.222 ±0.000[2]	0.053 ±0.000[5]	-0.018 ±0.000[6]	**0.225** ±0.000[1]	-0.606 ±0.000[8]	-0.057[7]	0.055 ±0.000[4]
ETT	-0.011 ±0.005	-0.399 ±0.212	**0.024** ±0.006[1,2]	-0.007 ±0.002[3,4]	-0.012 ±0.007[5–7]	-0.010 ±0.001[5–7]	-0.009 ±0.002[3,4]	**0.028** ±0.000[1,2]	-0.045[8]	-0.010 ±0.002[5–7]
Electric Devices	0.253 ±0.001	0.170 ±0.047	0.531 ±0.002[3]	0.645 ±0.001[2]	0.447 ±0.009[4]	**0.720** ±0.001[1]	0.216 ±0.006[7]	0.372 ±0.004[5]	-0.008[8]	0.304 ±0.008[6]
Favorita	-1.821 ±0.000	-1.713 ±0.001	-1.798 ±0.000[8]	-0.884 ±0.001[4]	-1.745 ±0.000[6]	-1.333 ±0.000[5]	**0.764** ±0.000[1]	-1.770 ±0.000[7]	-0.039[3]	0.759 ±0.001[2]
Avg. rank	–	–	3.7	3.8	4.0	4.2	4.3	5.0	5.2	5.7

TABLE S10

ANOMALY DETECTION: ABSOLUTE ROC-AUC OF EACH METHOD PER DATASET (HIGHER IS BETTER). CONVENTIONS ARE DESCRIBED IN SECTION S.4.1.

Anomaly	R-RNN	R-Tr	CoLES	Mask-InfoNCE	Slice-InfoNCE	JEPA	NTP-GRU	NTP-GPT	MLM	TSFresh
AGE	0.556 ±0.005	0.513 ±0.024	0.836 ±0.002[3]	0.911 ±0.003[2]	0.800 ±0.004[4]	**0.982** ±0.000[1]	0.745 ±0.002[5]	0.701 ±0.003[6]	0.604 ±0.010[7]	0.501[8]
AlphaBattle	0.645 ±0.025	0.668 ±0.013	**0.723** ±0.003[1]	0.702 ±0.009[2–6]	0.627 ±0.005[8]	0.696 ±0.014[2–6]	0.705 ±0.005[2–6]	0.701 ±0.019[2–6]	0.694 ±0.008[2–6]	0.658[7]
Retail	0.619 ±0.001	0.651 ±0.002	0.681 ±0.003[2,3]	**0.688** ±0.002[1]	0.680 ±0.006[2,3]	0.667 ±0.002[5]	0.674 ±0.002[4]	0.649 ±0.001[8]	0.664 ±0.002[6,7]	0.665[6,7]
Taobao	0.700 ±0.003	0.797 ±0.002	0.917 ±0.001[2,3]	0.904 ±0.000[4]	**0.930** ±0.001[1]	0.894 ±0.002[6]	0.900 ±0.001[5]	0.889 ±0.001[7]	0.919 ±0.002[2,3]	0.776[8]
Zvuk	0.803 ±0.002	0.794 ±0.000	0.918 ±0.000[2]	0.914 ±0.000[3]	**0.920** ±0.000[1]	0.841 ±0.003[8]	0.895 ±0.000[5]	0.899 ±0.001[4]	0.890 ±0.003[6]	0.877[7]
30Music	0.479 ±0.107	0.600 ±0.011	0.955 ±0.001[3]	**0.988** ±0.000[1]	0.840 ±0.001[5]	0.970 ±0.002[2]	0.726 ±0.005[7]	0.782 ±0.004[6]	0.946 ±0.001[4]	0.691[8]
Yambda	0.987 ±0.003	0.993 ±0.002	0.999 ±0.000[3–5]	**1.000** ±0.000[1,2]	**0.999** ±0.000[1,2]	0.999 ±0.001[3–5]	0.999 ±0.000[3–5]	0.999 ±0.000[6]	0.997 ±0.000[7]	0.991[8]
Rossmann	0.643 ±0.002	0.562 ±0.024	**0.904** ±0.001[1,2]	**0.907** ±0.002[1,2]	0.901 ±0.001[3]	0.882 ±0.001[4]	0.827 ±0.001[6]	0.849 ±0.000[5]	0.792 ±0.000[7]	0.452[8]
Favorita	0.750 ±0.331	0.500 ±0.217	**0.917** ±0.144[1–4]	0.792 ±0.191[5–7]	0.833 ±0.000[5–7]	**0.958** ±0.072[1–4]	**1.000** ±0.000[1–4]	**0.917** ±0.144[1–4]	0.667 ±0.577[5–7]	0.500[8]
Twitter	0.888 ±0.001	0.714 ±0.001	0.998 ±0.000[2]	0.996 ±0.000[6]	**0.999** ±0.000[1]	0.994 ±0.000[7]	0.998 ±0.000[3]	0.997 ±0.000[4]	0.996 ±0.000[5]	0.638[8]
Avg. rank	–	–	2.4	3.0	3.3	4.3	4.5	5.2	5.5	7.7

S.4.5 Embedding Fusion Analysis

Parent labels in the fusion tables indicate the configuration-selection task: R = regression, C = classification, F = forecasting, and A = anomaly detection.

Tables S13–S24 report the full AGE and Rossmann results for all six pairs of task-best encoders under both validators. For the CoLES–JEPA pair, CCA diverged under the linear probe on the two regression-type tasks ($R^2 < -10$); the affected entries are marked with a dash. On AGE, the preferred fusion strategy varies across encoder pairs, tasks, and validators. On Rossmann, the single-model baselines are already highly discriminative; consequently, the benefits of feature fusion are smaller, and TuckerFactorConcat, concatenation, and PCA perform similarly, with no single method clearly winning across all tasks. This is to be expected: despite its representational power, feature fusion constitutes a separate optimisation problem - the selection of the rank in the Tucker decomposition is NP-hard - and a learning problem that, like self-supervised learning, depends critically on the quality of the input embeddings; realising its full potential may require more careful hyperparameter tuning than simply fixing the rank to the minimum of the two feature dimensions.

Tables S25–S30 extend the fusion comparison to X5-Retail, using six pairs formed from the Slice-InfoNCE configuration selected for classification, Mask-InfoNCE selected for forecasting, and CoLES selected for regression or anomaly detection. Each table reports the two source encoders and their fused representations under a linear probe and LightGBM.

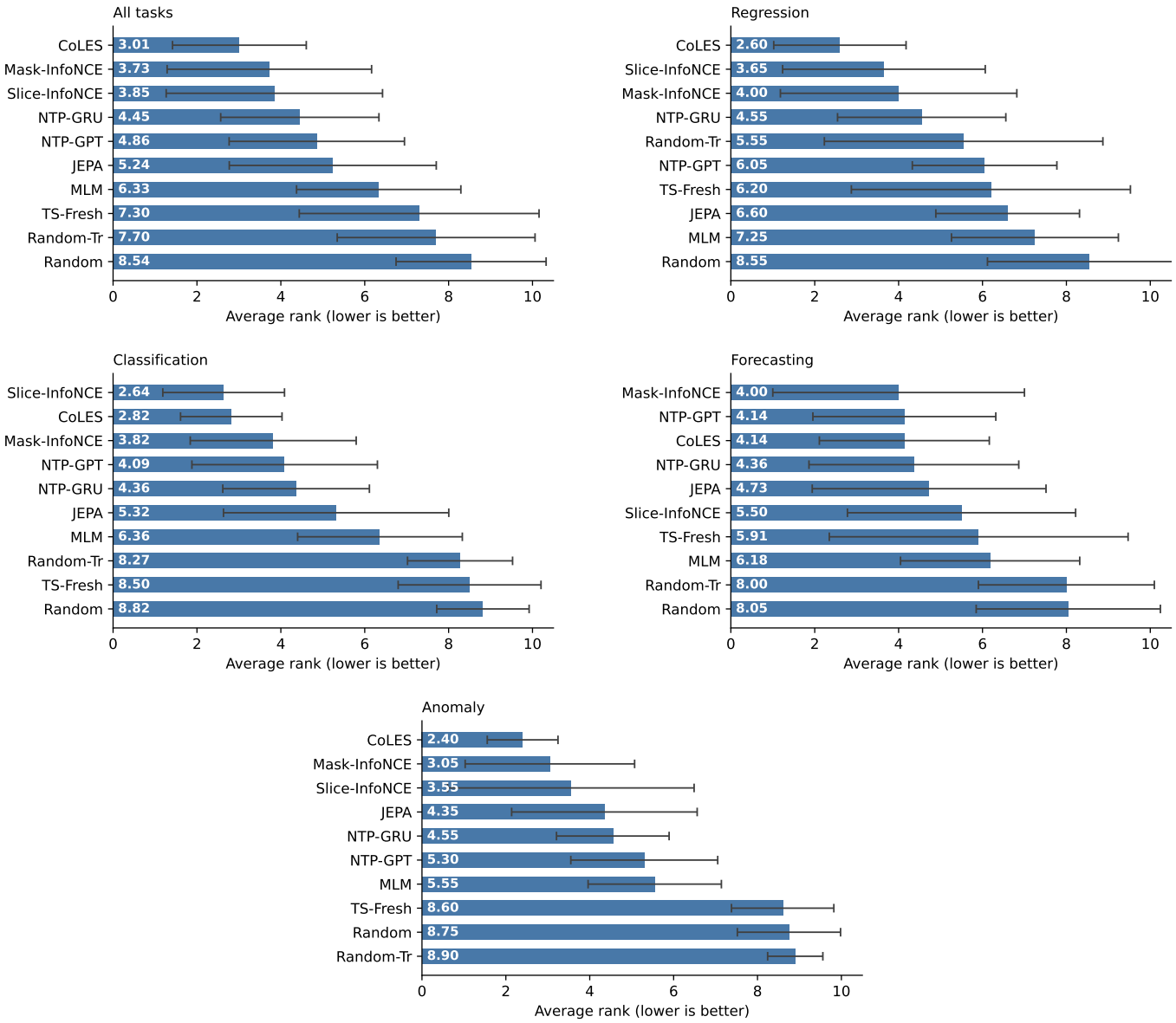


Fig. S3. Uncertainty-aware ranks of all ten methods (lower is better), shown as the mean and one standard deviation across all 42 dataset-task settings and separately for each task family. Methods are sorted by mean rank within each panel; ranks use the same mean-performance intervals and tie groups as Table S6.

S.4.6 Single- vs Multi-Objective Selection: Details

This section expands the configuration trade-off and HPO comparison of Section VI-E. Tables S31–S34 give the full per-configuration results underlying Table VI. Each row is one configuration: *Stop-by* indicates the downstream metric used to select it, and the four value columns report its performance on all tasks. The last row repeats the corresponding full-benchmark configuration for reference.

The best value for a task is not necessarily obtained by the configuration selected *for* that task: the diagonal entry (Stop-by equal to the evaluated task) is often not the column maximum. For instance, on X5-Retail with NTP-GRU the best regression score among the multi-target rows comes from the configuration selected by forecasting (0.121), not the one selected by regression (0.118). Since every candidate is validated on every task, we therefore report in Table VI the best value each regime attains per task (the column maximum), not the diagonal.

These results use a deliberately small sample, two datasets and two encoders, and should be read as an illustration rather than a full study; the reference rows show that the single- and multi-target numbers stay close to the full-benchmark configuration, so the small-sample comparison is not distorted.

S.5. Data Splitting Protocol

a) Global Time Split.: For datasets with local targets, we follow the Global Time Split of Time to Split [37]. We

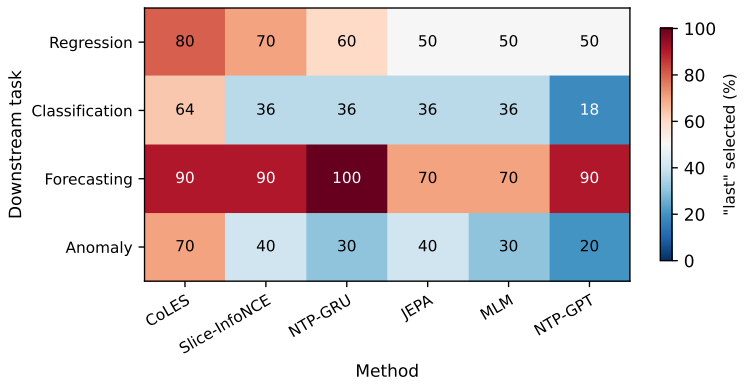


Fig. S4. Share of configurations in which the last hidden state is selected over mean-pooling, per task family and method.

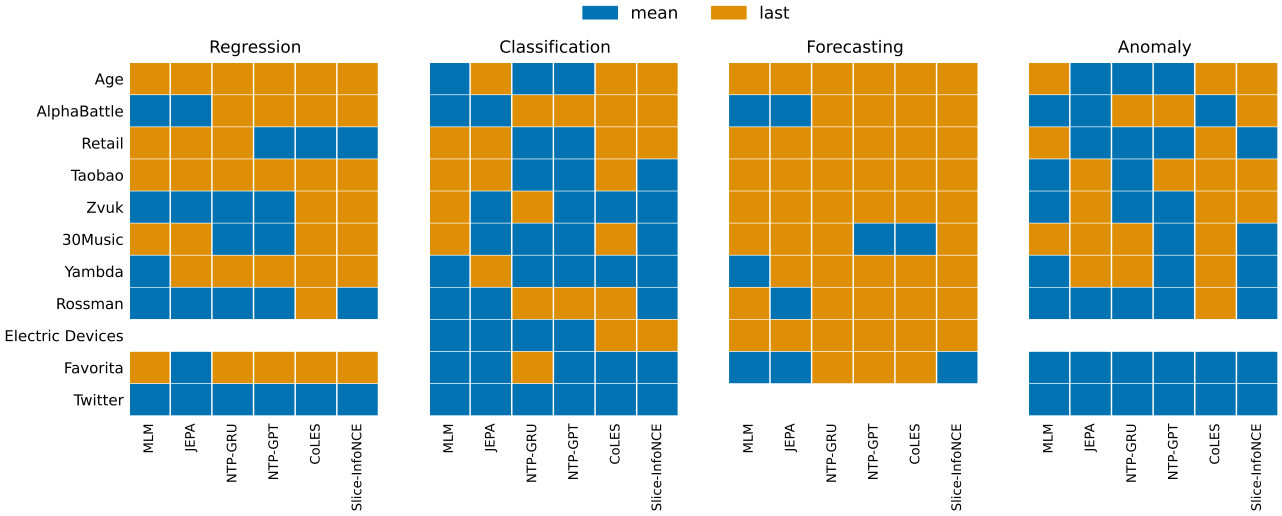


Fig. S5. Aggregation selected by hyperparameter optimisation for the dataset-method combinations included in this analysis, per task family. Blue = mean-pooling, orange = last hidden state; empty cells correspond to dataset-task combinations absent from the benchmark.

collect all event times that can serve as cutoffs with sufficient future context and define T_{test} as their global 0.9 quantile. This is one timestamp shared by all entities: events before it are available for training and validation, while events at or after it are reserved for testing, regardless of how many events each entity contributes on either side. Each dataset uses a horizon suited to its temporal resolution and activity scale; Table S1 reports the values used to construct local targets and exclude cutoffs without enough future context.

b) AlphaBattle source split.: AlphaBattle provides only relative within-client times rather than an absolute timeline shared across clients, so a dataset-wide T_{test} cannot be recovered. For its local targets, we therefore preserve the source-provided train and test partitions and sample cutoffs independently within them. Its global targets use the same entity-level split as the other datasets.

c) Global User Split.: Global targets, used mainly for classification and anomaly detection, describe an entire entity rather than the future after a cutoff. A temporal split is therefore unnecessary. Following the Global User Split of Time to Split [37], we assign complete entity histories to disjoint downstream training and test subsets in an approximately 90/10 proportion. The encoder and downstream model are trained on one set of entities and evaluated on unseen entities, each represented by an embedding of its full history. Datasets without explicit event time, such as Twitter, use this protocol directly. Exact processed counts are reported in Table S4.

d) Split metadata.: The Parquet rows contain the split information required by both target scopes. Temporal train and test directories encode either the global time boundary or, for AlphaBattle, the source-provided partition. The scalar `global_train` flag records the entity-level assignment used for global targets. For classification targets, entity splits are stratified when necessary to preserve class proportions and keep rare classes represented in each subset. Fixed seeds make both entity assignments and sampled shifts reproducible.

TABLE S11
PAIRWISE VALIDATOR CORRELATIONS BY DATASET FOR THE FIRST AND FINAL TEN OPTUNA TRIALS (LP IS A LINEAR PROBE).

Trials	Pair	AGE	Taobao	Twitter	Retail	Yambda	Zvuk
First 10	LightGBM–MLP	0.981	0.779	0.459	0.905	0.920	0.298
First 10	LP–LightGBM	0.976	0.765	0.226	0.864	0.870	0.451
First 10	LP–MLP	0.989	0.764	0.421	0.944	0.859	0.611
Final 10	LightGBM–MLP	0.900	0.667	0.494	0.970	0.758	0.581
Final 10	LP–LightGBM	0.917	0.513	0.495	0.944	0.764	0.554
Final 10	LP–MLP	0.928	0.680	0.918	0.970	0.684	0.595

TABLE S12
DETAILED FIRST-VERSUS-FINAL VALIDATOR RANKING SUMMARY. LOWER RANK AND REGRET ARE BETTER.

		First 10 Optuna trials			Final 10 Optuna trials		
Statistic	Scope	LightGBM	MLP	Linear probe	LightGBM	MLP	Linear probe
Winner	Total	10	7	6	11	5	7
Rank	Avg.	1.783	1.870	2.261	1.652	2.000	2.043
Rank	Cls.	1.833	1.500	2.500	1.167	2.167	2.500
Rank	Forecast	2.000	2.200	1.800	2.000	1.400	1.800
Rank	Reg.	1.000	2.333	2.667	1.333	2.333	2.333
Rank	Anomaly	2.333	1.500	2.000	2.167	2.000	1.500
Rank-S	Avg.	1.779	1.875	2.021	1.729	1.929	2.079
Regret	Avg.	0.334	1.075	0.665	0.345	0.971	0.840

TABLE S13
FUSION OF COLES (REGRESSION) AND NTP-GRU (FORECASTING) ON AGE UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS IN **BOLD**, THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	CoLES (R)	0.597 ±0.002	0.511 ±0.001	0.385 ±0.002	0.777 ±0.003
	NTP-GRU (F)	0.602 ±0.003	0.491 ±0.000	0.411 ±0.002	0.673 ±0.006
	CCA	0.590 ±0.003	0.510 ±0.002	0.412 ±0.001	0.764 ±0.001
	KrossFuse	0.608 ±0.001	0.500 ±0.001	0.407 ±0.002	0.749 ±0.006
	PCA	0.617 ±0.004	0.512 ±0.001	0.418 ±0.001	0.785 ±0.003
	TuckerFactorConcat	**0.620** ±0.002	**0.513** ±0.002	**0.420** ±0.000	**0.799** ±0.005
	Concatenation	0.610 ±0.005	0.511 ±0.003	0.419 ±0.001	0.780 ±0.006
LightGBM	CoLES (R)	0.599 ±0.004	**0.520** ±0.000	0.409 ±0.001	0.771 ±0.004
	NTP-GRU (F)	0.597 ±0.001	0.469 ±0.000	0.405 ±0.000	0.654 ±0.005
	CCA	0.581 ±0.003	0.496 ±0.000	0.408 ±0.000	0.749 ±0.009
	KrossFuse	0.611 ±0.003	0.500 ±0.000	0.406 ±0.001	0.715 ±0.003
	PCA	0.604 ±0.006	0.492 ±0.000	0.400 ±0.001	0.779 ±0.007
	TuckerFactorConcat	0.606 ±0.007	0.507 ±0.000	0.408 ±0.000	**0.800** ±0.006
	Concatenation	**0.615** ±0.004	0.518 ±0.000	**0.421** ±0.000	0.765 ±0.002

S.6. DOWNSTREAM VALIDATOR DETAILS

All downstream validators operate on frozen embeddings and are trained independently for each dataset–task pair. We do not tune validator hyperparameters: each validator uses the same fixed configuration across datasets, pretrained encoders, and embedding dimensions. Only the output dimension and learning objective depend on the downstream task; LightGBM additionally enables class balancing for anomaly detection.

a) Linear probe.: The linear probe is implemented as a single affine layer without hidden layers. Classification and anomaly-detection tasks use cross-entropy, whereas regression and forecasting tasks use mean-squared error. The probe is trained on a GPU with Adam, a learning rate of 3×10^{-4}, weight decay 2×10^{-6}, and batch size 512 for at most 200 epochs. We reserve 10% of the training examples for early stopping on the downstream validation metric, using a patience of 20 epochs; classification splits are stratified whenever possible. Final results are averaged over validator seeds 0, 1, and 2.

b) MLP.: The nonlinear probe has two hidden layers of widths 256 and 128 with ReLU activations and no dropout. It uses the same optimiser, batch size, validation split, and early-stopping procedure as the linear probe, with a maximum of 1,000 epochs.

c) LightGBM.: We use a fixed GBDT configuration with 1,000 trees, learning rate 0.02, maximum depth 12, 50 leaves,

TABLE S14

FUSION OF SLICE-INFONCE (CLASSIFICATION) AND JEPA (ANOMALY) ON AGE UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS IN **BOLD**, THE SECOND BEST IS UNDERLINED. DASHES MARK DIVERGED CCA RUNS ($R^2 < -10$).

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	Slice-InfoNCE (C)	**0.630** ±0.000	0.459 ±0.002	**0.244** ±0.001	0.745 ±0.002
	JEPA (A)	0.442 ±0.007	0.378 ±0.003	-0.011 ±0.008	0.982 ±0.000
	CCA	0.597 ±0.006	—	—	0.932 ±0.001
	KrossFuse	0.627 ±0.001	0.424 ±0.001	0.226 ±0.004	0.868 ±0.002
	PCA	0.623 ±0.003	**0.460** ±0.001	0.230 ±0.006	0.982 ±0.001
	TuckerFactorConcat	0.623 ±0.002	0.459 ±0.001	0.223 ±0.002	**0.988** ±0.000
	Concatenation	0.620 ±0.006	0.448 ±0.005	0.228 ±0.004	0.981 ±0.001
LightGBM	Slice-InfoNCE (C)	0.627 ±0.004	**0.472** ±0.001	**0.399** ±0.000	0.758 ±0.008
	JEPA (A)	0.486 ±0.002	0.376 ±0.001	-0.058 ±0.003	0.988 ±0.000
	CCA	0.591 ±0.006	0.419 ±0.000	0.102 ±0.001	0.946 ±0.003
	KrossFuse	0.622 ±0.007	0.431 ±0.000	0.237 ±0.001	0.820 ±0.001
	PCA	0.614 ±0.002	0.431 ±0.000	0.264 ±0.000	0.978 ±0.001
	TuckerFactorConcat	0.617 ±0.005	0.443 ±0.000	0.277 ±0.000	**0.992** ±0.000
	Concatenation	**0.628** ±0.005	0.461 ±0.001	0.386 ±0.001	0.987 ±0.000

TABLE S15

FUSION OF SLICE-INFONCE (CLASSIFICATION) AND NTP-GRU (FORECASTING) ON AGE UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS IN **BOLD**, THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	Slice-InfoNCE (C)	0.630 ±0.000	0.459 ±0.002	0.244 ±0.001	0.745 ±0.002
	NTP-GRU (F)	0.602 ±0.003	0.491 ±0.000	**0.411** ±0.002	0.673 ±0.006
	CCA	0.620 ±0.003	0.486 ±0.000	0.321 ±0.001	0.725 ±0.005
	KrossFuse	0.621 ±0.003	0.454 ±0.001	0.358 ±0.001	0.685 ±0.005
	PCA	0.627 ±0.002	0.494 ±0.001	0.404 ±0.000	**0.757** ±0.010
	TuckerFactorConcat	**0.631** ±0.002	0.497 ±0.001	**0.411** ±0.001	0.729 ±0.004
	Concatenation	0.630 ±0.002	**0.498** ±0.002	**0.411** ±0.001	0.741 ±0.006
LightGBM	Slice-InfoNCE (C)	0.627 ±0.004	0.472 ±0.001	0.399 ±0.000	0.758 ±0.008
	NTP-GRU (F)	0.597 ±0.001	0.469 ±0.000	0.405 ±0.000	0.654 ±0.005
	CCA	0.610 ±0.002	0.470 ±0.000	0.328 ±0.000	0.712 ±0.009
	KrossFuse	0.623 ±0.004	0.442 ±0.000	0.371 ±0.001	0.684 ±0.011
	PCA	0.623 ±0.003	0.471 ±0.000	0.394 ±0.000	0.757 ±0.005
	TuckerFactorConcat	0.624 ±0.002	0.481 ±0.000	0.403 ±0.001	0.718 ±0.010
	Concatenation	**0.629** ±0.004	**0.488** ±0.000	**0.422** ±0.000	**0.760** ±0.006

minimum 50 observations per leaf, row and feature subsampling rates of 0.75, and L_1 and L_2 regularisation coefficients equal to 1. Class balancing is enabled only for anomaly-detection tasks. The validator-sensitivity analysis uses random state 42 for all three validators.

S.7. UNSUPERVISED MODEL-SELECTION DETAILS

The SSL-loss analysis of Section VI-D compares hyperparameter configurations, whereas the following metrics are evaluated across checkpoints within a training run. Each metric is computed from frozen representations without downstream labels. Because their numerical ranges differ, the epoch-wise plots normalise each trajectory independently; checkpoint selection depends on the location of a curve's maximum rather than its scale.

a) RankMe.: RankMe is the effective rank of the singular-value distribution of a representation matrix. We compute it both on the final sequence embedding and immediately after categorical and numerical feature encoding, before the sequence backbone; the latter curve is denoted *first layer*.

b) LiDAR.: LiDAR computes a smooth rank from a discriminant covariance constructed using the SSL views as surrogate classes. *REG* in the figures denotes the diagonal regularisation added to the within-class covariance before inversion. We evaluate the displayed regularisation values because the resulting trajectory can be sensitive to this numerical choice.

TABLE S16

FUSION OF COLES (REGRESSION) AND SLICE-INFONCE (CLASSIFICATION) ON AGE UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS IN **BOLD**, THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	Slice-InfoNCE (C)	**0.630** ±0.000	0.459 ±0.002	0.244 ±0.001	0.745 ±0.002
	CoLES (R)	0.597 ±0.002	0.511 ±0.001	0.385 ±0.002	0.777 ±0.003
	CCA	0.611 ±0.002	0.489 ±0.001	0.306 ±0.001	0.803 ±0.004
	KrossFuse	0.624 ±0.007	0.486 ±0.001	0.267 ±0.000	0.731 ±0.005
	PCA	0.624 ±0.002	0.503 ±0.001	0.339 ±0.000	0.789 ±0.007
	TuckerFactorConcat	0.624 ±0.001	0.509 ±0.001	0.375 ±0.001	**0.810** ±0.002
	Concatenation	0.625 ±0.003	**0.512** ±0.001	**0.389** ±0.001	0.798 ±0.004
LightGBM	Slice-InfoNCE (C)	0.627 ±0.004	0.472 ±0.001	0.399 ±0.000	0.758 ±0.008
	CoLES (R)	0.599 ±0.004	**0.520** ±0.000	0.409 ±0.001	0.771 ±0.004
	CCA	0.603 ±0.005	0.478 ±0.000	0.322 ±0.001	0.788 ±0.005
	KrossFuse	0.624 ±0.004	0.494 ±0.000	0.278 ±0.000	0.704 ±0.004
	PCA	0.615 ±0.000	0.493 ±0.001	0.311 ±0.000	0.773 ±0.010
	TuckerFactorConcat	0.616 ±0.003	0.506 ±0.000	0.365 ±0.001	**0.812** ±0.008
	Concatenation	**0.629** ±0.004	0.519 ±0.001	**0.423** ±0.000	0.804 ±0.007

TABLE S17

FUSION OF COLES (REGRESSION) AND JEPA (ANOMALY) ON AGE UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS IN **BOLD**, THE SECOND BEST IS UNDERLINED. DASHES MARK DIVERGED CCA RUNS ($R^2 < -10$).

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	CoLES (R)	0.597 ±0.002	0.511 ±0.001	0.385 ±0.002	0.777 ±0.003
	JEPA (A)	0.442 ±0.007	0.378 ±0.003	-0.011 ±0.008	0.982 ±0.000
	CCA	0.581 ±0.003	—	—	0.966 ±0.003
	KrossFuse	0.592 ±0.002	0.501 ±0.003	0.361 ±0.003	0.974 ±0.001
	PCA	**0.605** ±0.001	**0.516** ±0.001	**0.387** ±0.002	0.991 ±0.000
	TuckerFactorConcat	0.605 ±0.004	0.513 ±0.002	0.364 ±0.005	**0.991** ±0.000
	Concatenation	0.588 ±0.001	0.507 ±0.002	0.370 ±0.002	0.984 ±0.000
LightGBM	CoLES (R)	0.599 ±0.004	**0.520** ±0.000	**0.409** ±0.001	0.771 ±0.004
	JEPA (A)	0.486 ±0.002	0.376 ±0.001	-0.058 ±0.003	0.988 ±0.000
	CCA	0.569 ±0.002	0.450 ±0.001	0.194 ±0.001	0.948 ±0.002
	KrossFuse	0.587 ±0.001	0.505 ±0.001	0.307 ±0.000	0.976 ±0.001
	PCA	0.593 ±0.005	0.498 ±0.001	0.316 ±0.001	0.981 ±0.000
	TuckerFactorConcat	0.587 ±0.004	0.500 ±0.000	0.349 ±0.000	**0.991** ±0.000
	Concatenation	**0.603** ±0.004	0.514 ±0.001	0.403 ±0.001	0.987 ±0.000

c) aSMI.: Aligned Sliced Mutual Information evaluates nonlinear dependence after projecting the representations. Its displayed parameters are the Gaussian-noise scale σ, projection dimension d, and number of nearest neighbours k used by the mutual-information estimator. We evaluate several combinations because no parameter choice can be selected from downstream labels in a label-free regime.

Figures S6–S9 provide additional trajectories for AGE and Twitter, NTP-GPT and CoLES, and classification, regression, and anomaly detection. Across these examples, changing the metric, representation layer, or metric parameters changes the selected epoch, and the unsupervised optimum does not consistently coincide with the downstream optimum.

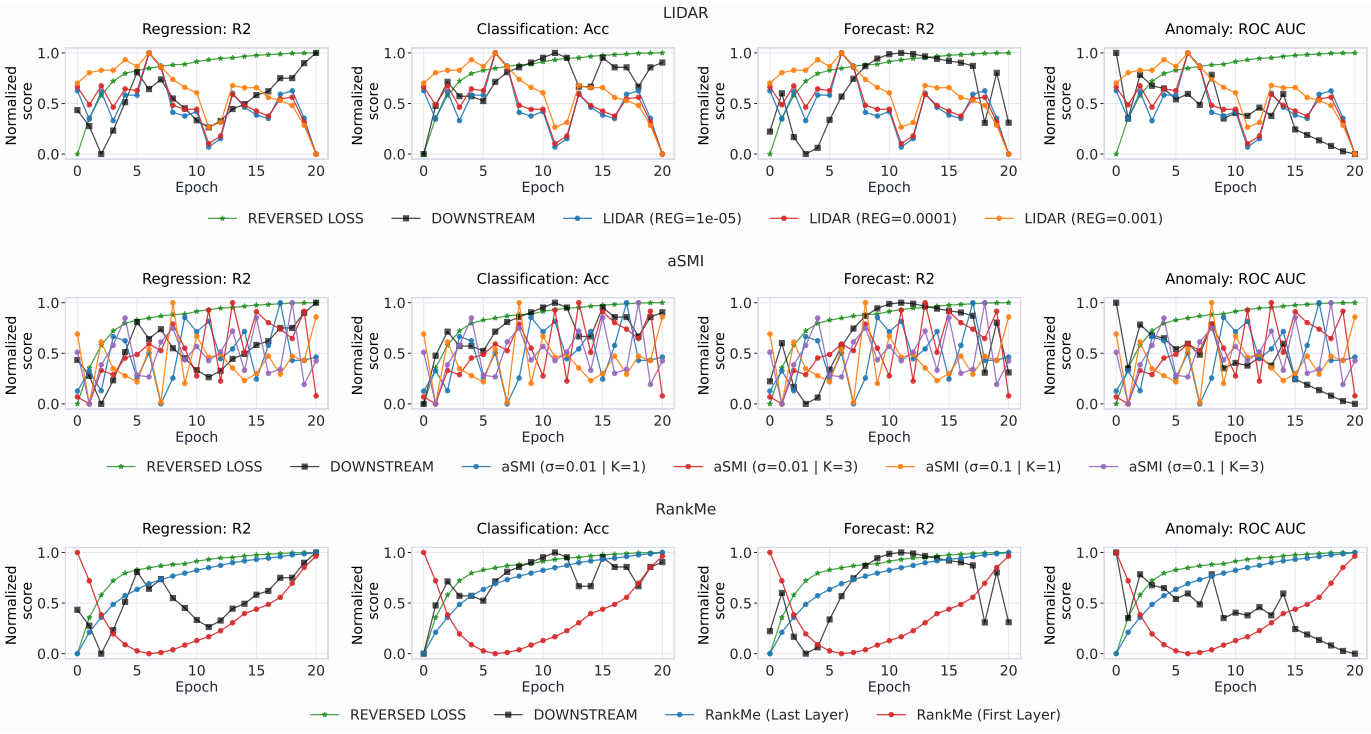


Fig. S6. NTP-GPT configuration selected for AGE anomaly detection: epoch-wise RankMe, LiDAR, and aSMI trajectories together with downstream performance and SSL loss.

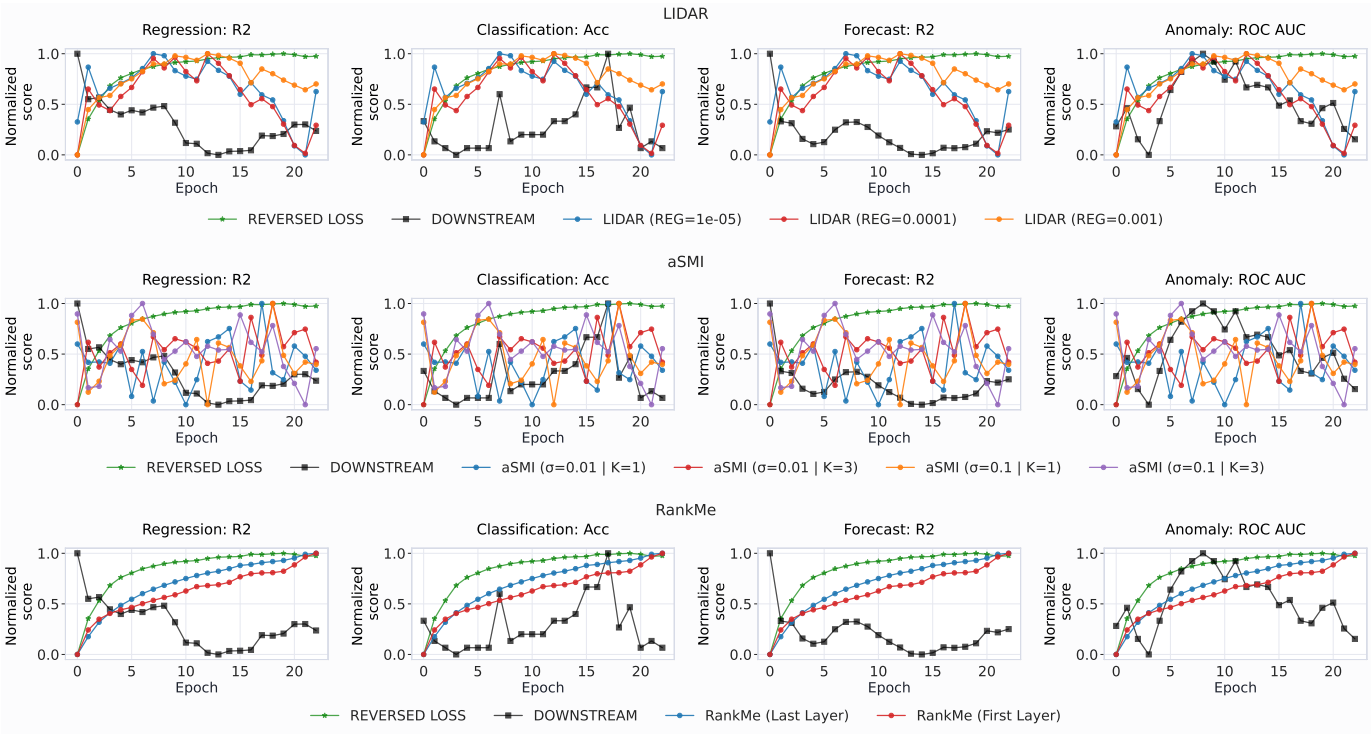


Fig. S7. NTP-GPT configuration selected for AGE classification: epoch-wise RankMe, LiDAR, and aSMI trajectories together with downstream performance and SSL loss.

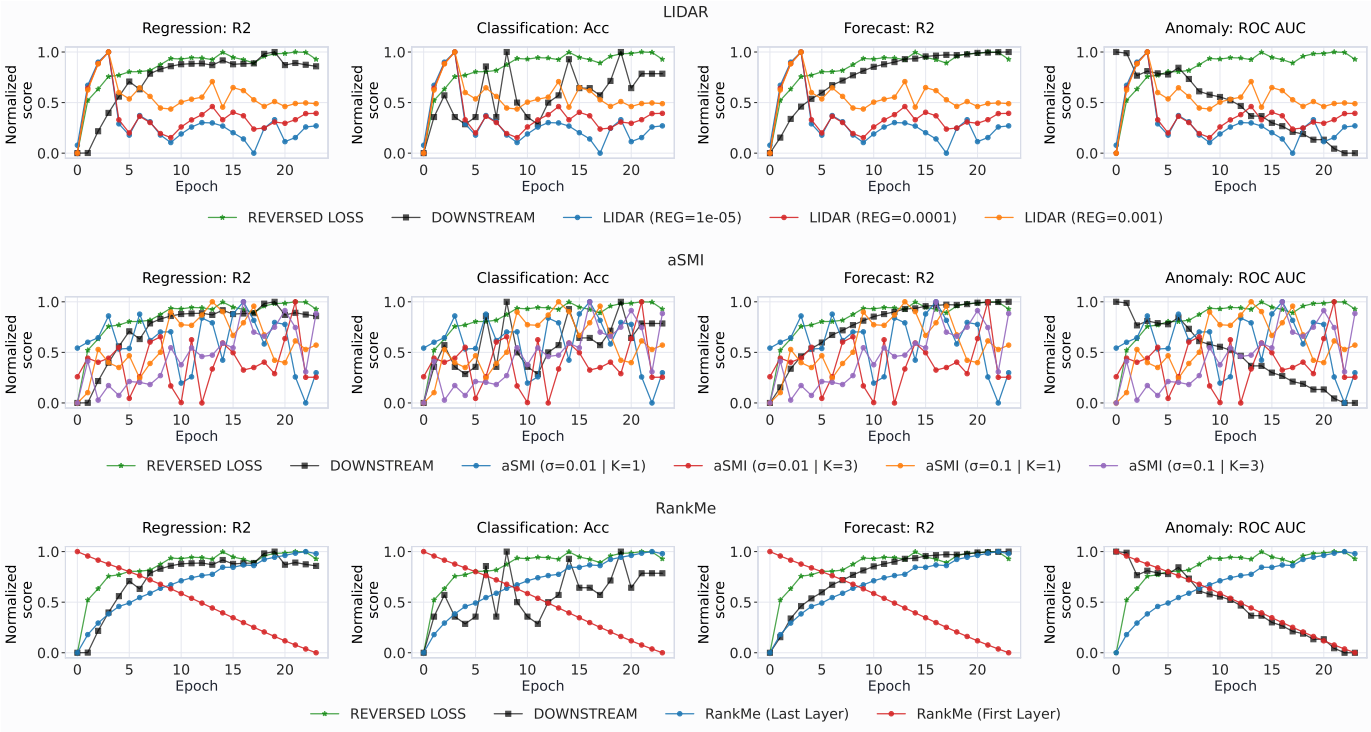


Fig. S8. CoLES configuration selected for AGE regression: epoch-wise RankMe, LiDAR, and aSMI trajectories together with downstream performance and SSL loss.

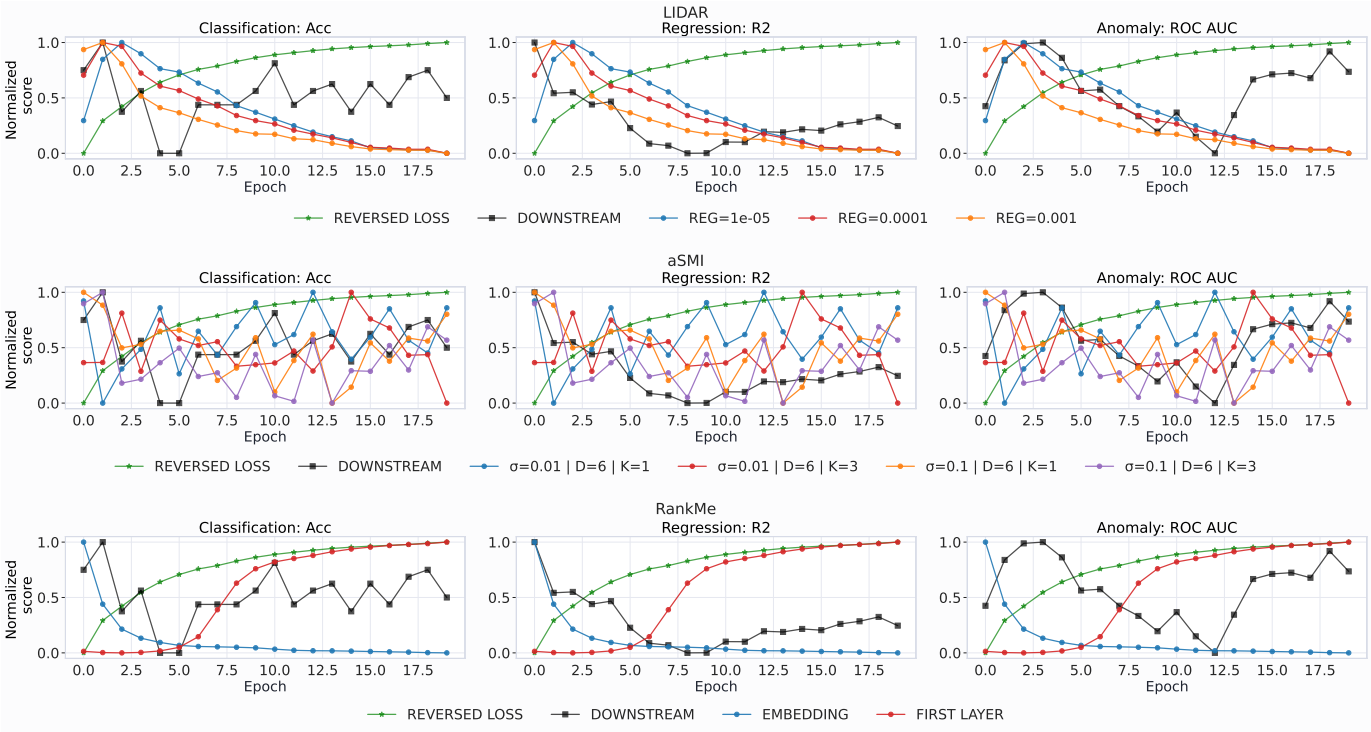


Fig. S9. NTP-GPT configuration selected for Twitter regression: epoch-wise RankMe, LiDAR, and aSMI trajectories together with downstream performance and SSL loss.

TABLE S18

FUSION OF NTP-GRU (FORECASTING) AND JEPA (ANOMALY) ON AGE UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS IN **BOLD**, THE SECOND BEST IS UNDERLINED. DASHES MARK DIVERGED CCA RUNS ($R^2 < -10$).

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	JEPA (A)	0.442 ±0.007	0.378 ±0.003	-0.011 ±0.008	0.982 ±0.000
	NTP-GRU (F)	0.602 ±0.003	0.491 ±0.000	0.411 ±0.002	0.673 ±0.006
	CCA	0.579 ±0.009	—	—	0.947 ±0.008
	KrossFuse	0.610 ±0.002	0.473 ±0.001	0.402 ±0.001	0.969 ±0.001
	PCA	**0.616** ±0.001	**0.499** ±0.001	**0.412** ±0.000	**0.991** ±0.001
	TuckerFactorConcat	**0.616** ±0.004	0.497 ±0.001	0.411 ±0.001	0.990 ±0.001
	Concatenation	0.614 ±0.004	0.492 ±0.000	0.410 ±0.001	0.985 ±0.000
LightGBM	JEPA (A)	0.486 ±0.002	0.376 ±0.001	-0.058 ±0.003	0.988 ±0.000
	NTP-GRU (F)	0.597 ±0.001	0.469 ±0.000	**0.405** ±0.000	0.654 ±0.005
	CCA	0.580 ±0.003	0.441 ±0.000	0.132 ±0.002	0.938 ±0.002
	KrossFuse	0.604 ±0.001	0.466 ±0.000	0.401 ±0.000	0.927 ±0.001
	PCA	0.605 ±0.004	0.470 ±0.001	0.397 ±0.000	0.980 ±0.000
	TuckerFactorConcat	0.602 ±0.005	0.468 ±0.000	0.397 ±0.001	**0.991** ±0.000
	Concatenation	**0.610** ±0.005	**0.476** ±0.000	0.404 ±0.000	0.987 ±0.000

TABLE S19

FUSION OF SLICE-INFONCE (CLASSIFICATION) AND COLES (ANOMALY) ON ROSSMANN UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS IN **BOLD**, THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	Slice-InfoNCE (C)	0.729 ±0.022	**0.951** ±0.000	-0.606 ±0.000	0.901 ±0.001
	CoLES (A)	0.688 ±0.000	0.945 ±0.003	0.053 ±0.000	**0.904** ±0.001
	CCA	0.170 ±0.009	-2.801 ±0.065	-2.018 ±0.003	0.659 ±0.003
	KrossFuse	**0.744** ±0.014	0.918 ±0.002	-0.915 ±0.000	0.882 ±0.001
	PCA	0.711 ±0.021	0.933 ±0.001	-2.049 ±0.000	0.865 ±0.002
	TuckerFactorConcat	0.735 ±0.014	0.932 ±0.001	-1.775 ±0.000	0.868 ±0.002
	Concatenation	0.720 ±0.021	0.936 ±0.002	**0.138** ±0.000	0.884 ±0.001
LightGBM	Slice-InfoNCE (C)	0.756 ±0.005	0.961 ±0.000	0.229 ±0.002	0.894 ±0.002
	CoLES (A)	**0.768** ±0.009	**0.970** ±0.001	**0.632** ±0.010	0.898 ±0.003
	CCA	0.577 ±0.014	0.175 ±0.028	-0.042 ±0.023	0.703 ±0.006
	KrossFuse	0.765 ±0.021	0.910 ±0.003	0.238 ±0.004	0.891 ±0.003
	PCA	0.726 ±0.005	0.944 ±0.001	0.248 ±0.002	0.879 ±0.004
	TuckerFactorConcat	0.726 ±0.005	0.956 ±0.000	0.247 ±0.001	0.891 ±0.002
	Concatenation	0.744 ±0.005	0.960 ±0.000	0.224 ±0.003	**0.899** ±0.002

TABLE S20

FUSION OF SLICE-INFONCE (CLASSIFICATION) AND JEPA (FORECASTING) ON ROSSMANN UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS IN **BOLD**, THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	Slice-InfoNCE (C)	0.729 ±0.022	0.951 ±0.000	-0.606 ±0.000	0.901 ±0.001
	JEPA (F)	0.488 ±0.065	0.920 ±0.001	**0.225** ±0.000	0.789 ±0.015
	CCA	0.223 ±0.039	0.651 ±0.008	-2.071 ±0.001	0.806 ±0.007
	KrossFuse	**0.750** ±0.015	0.937 ±0.003	0.052 ±0.001	0.886 ±0.005
	PCA	0.708 ±0.014	0.947 ±0.000	-2.041 ±0.000	0.882 ±0.001
	TuckerFactorConcat	0.690 ±0.111	0.947 ±0.000	-1.053 ±0.001	0.886 ±0.001
	Concatenation	0.747 ±0.019	**0.952** ±0.001	0.190 ±0.000	**0.902** ±0.002
LightGBM	Slice-InfoNCE (C)	0.756 ±0.005	**0.961** ±0.000	0.229 ±0.002	0.894 ±0.002
	JEPA (F)	0.741 ±0.024	0.924 ±0.000	0.225 ±0.001	0.802 ±0.003
	CCA	0.658 ±0.010	0.529 ±0.005	0.138 ±0.003	0.834 ±0.002
	KrossFuse	0.765 ±0.010	0.915 ±0.001	0.236 ±0.001	**0.896** ±0.001
	PCA	0.747 ±0.010	0.946 ±0.002	0.242 ±0.001	0.887 ±0.004
	TuckerFactorConcat	**0.768** ±0.009	0.953 ±0.000	**0.242** ±0.000	0.871 ±0.005
	Concatenation	0.747 ±0.005	0.958 ±0.000	0.230 ±0.002	0.894 ±0.002

TABLE S21

FUSION OF COLES (REGRESSION) AND SLICE-INFONCE (CLASSIFICATION) ON ROSSMANN UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS IN **BOLD**, THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	Slice-InfoNCE (C)	**0.729** ±0.022	0.951 ±0.000	-0.606 ±0.000	**0.901** ±0.001
	CoLES (R)	0.637 ±0.066	0.972 ±0.001	-1.032 ±0.000	0.747 ±0.001
	CCA	0.208 ±0.014	0.573 ±0.001	-2.062 ±0.000	0.722 ±0.012
	KrossFuse	0.583 ±0.055	0.933 ±0.001	-1.615 ±0.000	0.716 ±0.004
	PCA	0.604 ±0.054	0.964 ±0.000	-2.062 ±0.000	0.638 ±0.005
	TuckerFactorConcat	0.601 ±0.036	**0.974** ±0.000	-1.903 ±0.000	0.737 ±0.004
	Concatenation	0.688 ±0.018	0.966 ±0.000	**-0.163** ±0.000	0.713 ±0.007
LightGBM	Slice-InfoNCE (C)	0.756 ±0.005	0.961 ±0.000	0.229 ±0.002	**0.894** ±0.002
	CoLES (R)	0.747 ±0.010	**0.980** ±0.000	**0.845** ±0.001	0.867 ±0.001
	CCA	0.592 ±0.010	0.613 ±0.002	0.500 ±0.001	0.769 ±0.002
	KrossFuse	0.717 ±0.005	0.927 ±0.001	0.624 ±0.001	0.643 ±0.006
	PCA	0.717 ±0.014	0.867 ±0.004	0.621 ±0.003	0.583 ±0.004
	TuckerFactorConcat	0.738 ±0.005	0.967 ±0.001	0.817 ±0.002	0.785 ±0.002
	Concatenation	**0.759** ±0.009	0.966 ±0.001	0.843 ±0.001	0.706 ±0.006

TABLE S22

FUSION OF COLES (REGRESSION) AND COLES (ANOMALY) ON ROSSMANN UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS IN **BOLD**, THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	CoLES (A)	0.688 ±0.000	0.945 ±0.003	0.053 ±0.000	0.904 ±0.001
	CoLES (R)	0.637 ±0.066	**0.972** ±0.001	-1.032 ±0.000	0.747 ±0.001
	CCA	0.262 ±0.031	0.938 ±0.001	-2.055 ±0.002	0.881 ±0.005
	KrossFuse	0.682 ±0.019	0.959 ±0.001	-1.059 ±0.000	0.898 ±0.001
	PCA	0.676 ±0.029	0.918 ±0.002	-2.056 ±0.000	**0.920** ±0.001
	TuckerFactorConcat	0.598 ±0.086	0.936 ±0.001	-1.818 ±0.000	0.920 ±0.001
	Concatenation	**0.702** ±0.021	0.935 ±0.002	**0.097** ±0.000	0.912 ±0.002
LightGBM	CoLES (A)	0.768 ±0.009	0.970 ±0.001	0.632 ±0.010	0.898 ±0.003
	CoLES (R)	0.747 ±0.010	**0.980** ±0.000	**0.845** ±0.001	0.867 ±0.001
	CCA	0.637 ±0.005	0.970 ±0.000	0.758 ±0.001	0.874 ±0.006
	KrossFuse	0.762 ±0.010	0.975 ±0.000	0.776 ±0.001	0.904 ±0.000
	PCA	0.756 ±0.010	0.956 ±0.001	0.725 ±0.001	**0.909** ±0.002
	TuckerFactorConcat	0.756 ±0.005	0.958 ±0.000	0.818 ±0.002	0.904 ±0.001
	Concatenation	**0.771** ±0.014	0.971 ±0.000	0.740 ±0.003	0.904 ±0.003

TABLE S23

FUSION OF COLES (REGRESSION) AND JEPA (FORECASTING) ON ROSSMANN UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS IN **BOLD**, THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	CoLES (R)	0.637 ±0.066	0.972 ±0.001	-1.032 ±0.000	0.747 ±0.001
	JEPA (F)	0.488 ±0.065	0.920 ±0.001	0.225 ±0.000	0.789 ±0.015
	CCA	0.253 ±0.068	0.971 ±0.000	-2.057 ±0.001	0.882 ±0.004
	KrossFuse	**0.702** ±0.005	**0.977** ±0.001	0.050 ±0.000	0.787 ±0.003
	PCA	0.304 ±0.082	0.924 ±0.001	-1.995 ±0.000	**0.892** ±0.001
	TuckerFactorConcat	0.571 ±0.015	0.951 ±0.001	-1.097 ±0.001	0.887 ±0.001
	Concatenation	0.693 ±0.010	0.974 ±0.000	**0.233** ±0.000	0.766 ±0.002
LightGBM	CoLES (R)	0.747 ±0.010	**0.980** ±0.000	0.845 ±0.001	0.867 ±0.001
	JEPA (F)	0.741 ±0.024	0.924 ±0.000	0.225 ±0.001	0.802 ±0.003
	CCA	0.649 ±0.022	0.977 ±0.000	0.629 ±0.002	**0.898** ±0.003
	KrossFuse	0.750 ±0.009	0.975 ±0.000	0.805 ±0.001	0.841 ±0.002
	PCA	**0.774** ±0.014	0.944 ±0.001	0.776 ±0.001	0.874 ±0.003
	TuckerFactorConcat	0.750 ±0.009	0.961 ±0.000	0.824 ±0.001	0.881 ±0.004
	Concatenation	0.753 ±0.005	0.976 ±0.000	**0.846** ±0.001	0.861 ±0.002

TABLE S24

FUSION OF JEPA (FORECASTING) AND COLES (ANOMALY) ON ROSSMANN UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS IN **BOLD**, THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	CoLES (A)	0.688 ±0.000	0.945 ±0.003	0.053 ±0.000	0.904 ±0.001
	JEPA (F)	0.488 ±0.065	0.920 ±0.001	0.225 ±0.000	0.789 ±0.015
	CCA	0.244 ±0.059	**0.954** ±0.003	-2.047 ±0.002	0.824 ±0.026
	KrossFuse	**0.723** ±0.009	0.952 ±0.004	**0.241** ±0.000	0.906 ±0.000
	PCA	0.637 ±0.014	0.939 ±0.003	-1.989 ±0.000	**0.917** ±0.002
	TuckerFactorConcat	0.625 ±0.101	0.944 ±0.002	-0.983 ±0.001	0.917 ±0.002
	Concatenation	0.702 ±0.027	0.936 ±0.002	0.138 ±0.000	0.884 ±0.001
LightGBM	CoLES (A)	**0.768** ±0.009	**0.970** ±0.001	0.632 ±0.010	0.898 ±0.003
	JEPA (F)	0.741 ±0.024	0.924 ±0.000	0.225 ±0.001	0.802 ±0.003
	CCA	0.565 ±0.026	0.757 ±0.154	0.406 ±0.032	0.872 ±0.005
	KrossFuse	0.753 ±0.005	0.966 ±0.000	**0.691** ±0.000	**0.913** ±0.001
	PCA	0.750 ±0.018	0.957 ±0.000	0.661 ±0.003	0.898 ±0.003
	TuckerFactorConcat	0.741 ±0.000	0.956 ±0.000	0.670 ±0.001	0.892 ±0.002
	Concatenation	0.750 ±0.015	0.960 ±0.000	0.224 ±0.003	0.899 ±0.002

TABLE S25

FUSION OF SLICE-INFONCE (CLASSIFICATION-SELECTED) AND MASK-INFONCE (FORECASTING-SELECTED) ON X5-RETAIL UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS BOLD; THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	Slice-InfoNCE (C)	0.517 ±0.005	0.145 ±0.002	0.309 ±0.007	**0.680** ±0.006
	Mask-InfoNCE (F)	0.312 ±0.003	0.045 ±0.006	**0.371** ±0.001	0.643 ±0.001
	Concatenation	0.511 ±0.000	**0.169** ±0.008	0.222 ±0.069	0.665 ±0.011
	PCA	**0.517** ±0.001	0.169 ±0.003	0.314 ±0.007	0.679 ±0.001
	CCA	0.442 ±0.002	0.007 ±0.001	-0.074 ±0.005	0.607 ±0.004
	TuckerFactorConcat	0.514 ±0.002	0.166 ±0.001	0.319 ±0.005	0.679 ±0.001
	KrossFuse	0.448 ±0.003	0.134 ±0.001	0.091 ±0.004	0.613 ±0.003
LightGBM	Slice-InfoNCE (C)	**0.525** ±0.002	0.158 ±0.001	**0.481** ±0.002	**0.677** ±0.000
	Mask-InfoNCE (F)	0.325 ±0.001	0.061 ±0.000	0.374 ±0.000	0.625 ±0.001
	Concatenation	0.519 ±0.006	**0.171** ±0.000	0.272 ±0.183	0.674 ±0.017
	PCA	0.506 ±0.001	0.170 ±0.001	0.300 ±0.001	0.663 ±0.004
	CCA	0.432 ±0.006	0.073 ±0.000	0.066 ±0.000	0.584 ±0.013
	TuckerFactorConcat	0.511 ±0.003	0.170 ±0.000	0.333 ±0.001	0.666 ±0.005
	KrossFuse	0.480 ±0.004	0.161 ±0.000	0.199 ±0.001	0.636 ±0.003

TABLE S26

FUSION OF SLICE-INFONCE (CLASSIFICATION-SELECTED) AND COLES (ANOMALY-SELECTED) ON X5-RETAIL UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS BOLD; THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	Slice-InfoNCE (C)	0.517 ±0.005	0.145 ±0.002	0.309 ±0.007	0.680 ±0.006
	CoLES (A)	0.502 ±0.005	0.052 ±0.004	0.271 ±0.003	0.681 ±0.003
	Concatenation	0.520 ±0.001	0.141 ±0.008	0.360 ±0.007	0.678 ±0.002
	PCA	0.516 ±0.002	0.146 ±0.002	0.321 ±0.009	**0.686** ±0.002
	CCA	0.513 ±0.002	0.104 ±0.004	0.330 ±0.005	0.681 ±0.002
	TuckerFactorConcat	**0.520** ±0.002	0.139 ±0.001	**0.365** ±0.002	0.683 ±0.000
	KrossFuse	0.511 ±0.003	**0.153** ±0.001	0.259 ±0.005	0.674 ±0.001
LightGBM	Slice-InfoNCE (C)	**0.525** ±0.002	**0.158** ±0.001	0.481 ±0.002	0.677 ±0.000
	CoLES (A)	0.498 ±0.001	0.051 ±0.000	0.371 ±0.001	0.674 ±0.003
	Concatenation	0.517 ±0.004	0.155 ±0.000	**0.530** ±0.001	0.679 ±0.001
	PCA	0.512 ±0.002	0.139 ±0.001	0.324 ±0.000	**0.682** ±0.003
	CCA	0.497 ±0.001	0.112 ±0.000	0.269 ±0.001	0.673 ±0.003
	TuckerFactorConcat	0.515 ±0.003	0.141 ±0.001	0.359 ±0.001	0.680 ±0.003
	KrossFuse	0.501 ±0.003	0.147 ±0.000	0.271 ±0.000	0.676 ±0.002

TABLE S27

FUSION OF MASK-INFONCE (FORECASTING-SELECTED) AND COLES (ANOMALY-SELECTED) ON X5-RETAIL UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS BOLD; THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	Mask-InfoNCE (F)	0.312 ±0.003	0.045 ±0.006	0.371 ±0.001	0.643 ±0.001
	CoLES (A)	**0.502** ±0.005	0.052 ±0.004	0.271 ±0.003	**0.681** ±0.003
	Concatenation	0.312 ±0.009	**0.143** ±0.002	**0.383** ±0.007	0.642 ±0.001
	PCA	0.318 ±0.004	0.137 ±0.003	0.309 ±0.004	0.640 ±0.004
	CCA	0.295 ±0.001	0.049 ±0.001	0.110 ±0.001	0.575 ±0.003
	TuckerFactorConcat	0.309 ±0.005	0.133 ±0.001	0.270 ±0.002	0.635 ±0.006
	KrossFuse	0.300 ±0.006	0.107 ±0.002	0.187 ±0.002	0.614 ±0.003
LightGBM	Mask-InfoNCE (F)	0.325 ±0.001	0.061 ±0.000	0.374 ±0.000	0.625 ±0.001
	CoLES (A)	**0.498** ±0.001	0.051 ±0.000	0.371 ±0.001	**0.674** ±0.003
	Concatenation	0.323 ±0.004	**0.123** ±0.000	**0.379** ±0.001	0.623 ±0.003
	PCA	0.304 ±0.002	0.119 ±0.000	0.326 ±0.001	0.624 ±0.001
	CCA	0.299 ±0.003	0.112 ±0.001	0.167 ±0.000	0.581 ±0.008
	TuckerFactorConcat	0.300 ±0.004	0.122 ±0.000	0.304 ±0.001	0.626 ±0.002
	KrossFuse	0.300 ±0.002	0.112 ±0.000	0.211 ±0.001	0.614 ±0.002

TABLE S28

FUSION OF COLES (REGRESSION-SELECTED) AND SLICE-INFONCE (CLASSIFICATION-SELECTED) ON X5-RETAIL UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS BOLD; THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	Slice-InfoNCE (C)	**0.517** ±0.005	0.145 ±0.002	**0.309** ±0.007	**0.680** ±0.006
	CoLES (R)	0.513 ±0.005	0.159 ±0.004	0.170 ±0.005	0.676 ±0.001
	Concatenation	0.507 ±0.003	**0.192** ±0.001	0.193 ±0.003	0.663 ±0.004
	PCA	0.496 ±0.004	0.146 ±0.000	0.129 ±0.002	0.658 ±0.003
	CCA	0.385 ±0.006	0.033 ±0.002	-0.016 ±0.002	0.580 ±0.003
	TuckerFactorConcat	0.515 ±0.003	0.192 ±0.001	0.158 ±0.005	0.670 ±0.002
	KrossFuse	0.466 ±0.003	0.026 ±0.002	0.084 ±0.001	0.611 ±0.003
LightGBM	Slice-InfoNCE (C)	**0.525** ±0.002	0.158 ±0.001	**0.481** ±0.002	**0.677** ±0.000
	CoLES (R)	0.515 ±0.002	0.169 ±0.000	0.165 ±0.000	0.658 ±0.005
	Concatenation	0.507 ±0.001	0.188 ±0.001	0.173 ±0.001	0.651 ±0.003
	PCA	0.501 ±0.003	0.152 ±0.001	0.151 ±0.000	0.632 ±0.004
	CCA	0.383 ±0.002	0.062 ±0.000	0.039 ±0.000	0.571 ±0.007
	TuckerFactorConcat	0.507 ±0.003	**0.194** ±0.001	0.158 ±0.001	0.658 ±0.006
	KrossFuse	0.479 ±0.003	0.091 ±0.001	0.128 ±0.001	0.628 ±0.005

TABLE S29

FUSION OF COLES (REGRESSION-SELECTED) AND MASK-INFONCE (FORECASTING-SELECTED) ON X5-RETAIL UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS BOLD; THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	Mask-InfoNCE (F)	0.312 ±0.003	0.045 ±0.006	**0.371** ±0.001	0.643 ±0.001
	CoLES (R)	**0.513** ±0.005	0.159 ±0.004	0.170 ±0.005	**0.676** ±0.001
	Concatenation	0.506 ±0.004	**0.174** ±0.001	0.179 ±0.015	0.664 ±0.008
	PCA	0.504 ±0.002	0.173 ±0.001	0.162 ±0.001	0.654 ±0.003
	CCA	0.442 ±0.002	0.022 ±0.002	-0.129 ±0.002	0.581 ±0.010
	TuckerFactorConcat	0.505 ±0.003	0.167 ±0.001	0.175 ±0.003	0.655 ±0.002
	KrossFuse	0.474 ±0.002	0.103 ±0.002	0.098 ±0.001	0.609 ±0.002
LightGBM	Mask-InfoNCE (F)	0.325 ±0.001	0.061 ±0.000	**0.374** ±0.000	0.625 ±0.001
	CoLES (R)	**0.515** ±0.002	0.169 ±0.000	0.165 ±0.000	**0.658** ±0.005
	Concatenation	0.511 ±0.003	**0.170** ±0.001	0.166 ±0.000	0.655 ±0.001
	PCA	0.494 ±0.006	0.147 ±0.000	0.150 ±0.001	0.631 ±0.009
	CCA	0.414 ±0.006	0.044 ±0.001	0.021 ±0.000	0.577 ±0.008
	TuckerFactorConcat	0.504 ±0.002	0.166 ±0.001	0.164 ±0.000	0.655 ±0.002
	KrossFuse	0.488 ±0.007	0.141 ±0.000	0.142 ±0.000	0.646 ±0.004

TABLE S30

FUSION OF COLES (REGRESSION-SELECTED) AND COLES (ANOMALY-SELECTED) ON X5-RETAIL UNDER BOTH VALIDATORS. VALUES ARE MEAN ± STANDARD DEVIATION ACROSS REPEATED DOWNSTREAM RUNS. THE BEST MEAN PER COLUMN WITHIN EACH VALIDATOR BLOCK IS BOLD; THE SECOND BEST IS UNDERLINED.

		Classification accuracy	Regression R^2	Forecasting R^2	Anomaly ROC-AUC
Linear probe	CoLES (A)	0.502 ±0.005	0.052 ±0.004	**0.271** ±0.003	**0.681** ±0.003
	CoLES (R)	0.513 ±0.005	0.159 ±0.004	0.170 ±0.005	<u>0.676 ±0.001</u>
	Concatenation	0.513 ±0.002	<u>0.183 ±0.005</u>	<u>0.178 ±0.009</u>	0.674 ±0.001
	PCA	**0.518** ±0.002	**0.184** ±0.001	0.136 ±0.004	0.673 ±0.001
	CCA	0.394 ±0.004	0.023 ±0.001	0.003 ±0.004	0.576 ±0.005
	TuckerFactorConcat	<u>0.516 ±0.002</u>	0.179 ±0.002	0.138 ±0.004	0.667 ±0.001
	KrossFuse	0.498 ±0.005	0.170 ±0.001	0.135 ±0.001	0.658 ±0.002
LightGBM	CoLES (A)	0.498 ±0.001	0.051 ±0.000	**0.371** ±0.001	**0.674** ±0.003
	CoLES (R)	**0.515** ±0.002	0.169 ±0.000	0.165 ±0.000	<u>0.658 ±0.005</u>
	Concatenation	<u>0.514 ±0.002</u>	**0.184** ±0.000	<u>0.168 ±0.001</u>	0.654 ±0.007
	PCA	0.509 ±0.001	0.180 ±0.001	0.153 ±0.000	0.653 ±0.005
	CCA	0.397 ±0.004	0.065 ±0.001	0.037 ±0.001	0.563 ±0.013
	TuckerFactorConcat	0.506 ±0.002	<u>0.181 ±0.001</u>	0.152 ±0.001	0.652 ±0.004
	KrossFuse	0.499 ±0.003	0.172 ±0.001	0.150 ±0.001	0.637 ±0.006

TABLE S31
X5-RETAIL, NTP-GRU: SINGLE- VS MULTI-TARGET SELECTION.

HPO	Stop-by	Reg.	Clf.	Forec.	Anom.
Single-target HPO					
Single	Reg.	0.160	0.417	0.395	0.668
Single	Clf.	0.097	0.454	0.156	0.658
Single	Forec.	0.060	0.442	0.431	0.650
Single	Anom.	0.082	0.435	0.132	0.683
Multi-target HPO					
Multi	Reg.	0.118	0.339	0.348	0.640
Multi	Clf.	0.103	0.443	0.133	0.658
Multi	Forec.	0.121	0.426	0.428	0.648
Multi	Anom.	0.076	0.407	0.140	0.663
Full benchmark (ref.)		0.138	0.463	0.412	0.674

TABLE S32
X5-RETAIL, JEPA: SINGLE- VS MULTI-TARGET SELECTION.

HPO	Stop-by	Reg.	Clf.	Forec.	Anom.
Single-target HPO					
Single	Reg.	0.125	0.305	0.059	0.639
Single	Clf.	-0.053	0.463	0.075	0.673
Single	Forec.	0.044	0.392	0.325	0.651
Single	Anom.	0.070	0.322	0.094	0.691
Multi-target HPO					
Multi	Reg.	0.146	0.355	0.272	0.660
Multi	Clf.	0.063	0.434	0.185	0.591
Multi	Forec.	0.091	0.441	0.336	0.657
Multi	Anom.	0.041	0.427	0.132	0.692
Full benchmark (ref.)		0.062	0.479	0.403	0.667

TABLE S33
TWITTER, NTP-GRU: SINGLE- VS MULTI-TARGET SELECTION.

HPO	Stop-by	Reg.	Clf.	Forec.	Anom.
Single-target HPO					
Single	Reg.	0.685	0.718	–	0.995
Single	Clf.	0.626	0.744	–	0.993
Single	Anom.	0.566	0.700	–	0.997
Multi-target HPO					
Multi	Reg.	0.677	0.729	–	0.995
Multi	Clf.	0.670	0.751	–	0.996
Multi	Anom.	0.666	0.731	–	0.997
Full benchmark (ref.)		0.689	0.748	–	0.998

TABLE S34
TWITTER, JEPA: SINGLE- VS MULTI-TARGET SELECTION.

HPO	Stop-by	Reg.	Clf.	Forec.	Anom.
Single-target HPO					
Single	Reg.	0.389	0.584	–	0.803
Single	Clf.	0.219	0.601	–	0.929
Single	Anom.	0.450	0.647	–	0.998
Multi-target HPO					
Multi	Reg.	0.480	0.634	–	0.994
Multi	Clf.	0.477	0.646	–	0.991
Multi	Anom.	0.461	0.630	–	0.994
Full benchmark (ref.)		0.498	0.635	–	0.999